Silicon Validation Engineer

Electronics & Programming
A Textbook-Style Guide: Concepts, Types, Formulae and Solved Problems with Diagrams

Every problem is shown as **Question → Diagram → Step-by-step Solution → Answer**

How to use this book
1. Read the short explanation and the **types** tables of a chapter. 2. Learn the yellow **formula box**. 3. For each problem, read the question and study the diagram, **try it yourself**, then compare with the solution below it. 4. Finish with the quick questions and the one-page cheat sheet (Chapter 14). *Conventions: $V_T = kT/q \approx 25.85$ mV at 300 K; mains = 50 Hz; silicon diode/BJT drop = 0.7 V.*

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1. Linear Components – Resistor, Capacitor, Inductor

In one line: R, C and L are the three basic *linear* passive components. A component is *linear* when doubling the input (voltage) doubles the output (current). R reacts instantly; C and L store energy, so their response depends on **time** and **frequency**.

1.1 Basics at a glance

	Resistor (R)	Capacitor (C)	Inductor (L)
Unit	Ohm (Ω)	Farad (F): pF, nF, µF	Henry (H): nH, µH, mH
V–I law	$v = i \cdot R$	$i = C \cdot dv/dt$	$v = L \cdot di/dt$
Phase of current (AC)	In phase (0°)	**Leads** voltage by 90°	**Lags** voltage by 90°
Impedance	R	$X_C = 1/(2\pi fC)$	$X_L = 2\pi fL$
DC (steady state)	Passes, drops V = IR	**Open circuit** (blocks DC)	**Short circuit** (passes DC)
High frequency	Unchanged	Acts like a short	Acts like an open
Energy	Dissipated as heat	Stored: $\frac{1}{2}CV^2$	Stored: $\frac{1}{2}LI^2$
Cannot change instantly	—	Voltage across C	Current through L

Memory trick – "ELI the ICE man": in an inductor (L) voltage **E** leads current **I**; in a capacitor (C) current **I** leads voltage **E**.

1.2 Types of components and where each is used

Component	Types	Typical use
Resistor	Carbon-film, metal-film (precise, low noise), wire-wound (high power), SMD chip, variable (potentiometer/trimmer), thermistor (NTC/PTC), LDR	Current limiting, dividers, pull-up/down, sensing; thermistor for temperature, LDR for light
Capacitor	Ceramic (small, non-polar, high f), electrolytic (large, polarised), tantalum, film (stable, audio), supercapacitor, variable	Decoupling/bypass, filtering, timing, coupling (blocks DC), energy storage
Inductor	Air-core, ferrite-core, iron-core, toroid, SMD power inductor, chokes, transformer (coupled inductors)	Switching converters, filters, RF tuning, EMI chokes

1.3 Time response – the time constant τ

When a step voltage is applied, a capacitor charges (or an inductor current rises) **exponentially**. After 1τ the quantity has reached 63.2 % of its final value, after 5τ it is effectively complete (99.3 %). For RC: $\tau = RC$; for RL: $\tau = L/R$.

Key formulae	
RC charging	$v_C(t) = V_s(1 - e^{-t/\tau})$, discharging: $v_C(t) = V_0 e^{-t/\tau}$
RL current rise	$i(t) = (V/R)(1 - e^{-t/\tau})$
Time to reach x %	$t = -\tau \ln(1 - x/100)$: 63.2 % → 1τ; 90 % → 2.3τ; 99 % → 4.6τ
Rise time (10–90 %)	$t_r = 2.2\tau$
Series / parallel	R and L: series add, parallel $1/R_p = \Sigma 1/R$. C: parallel add, series $1/C_s = \Sigma 1/C$
Resonance (series RLC)	$f_0 = 1/(2\pi\sqrt{LC})$; $Q = X_L/R = (1/R)\sqrt{L/C}$; $BW = f_0/Q$
Impedance magnitude	$\lvert Z\rvert = \sqrt{R^2 + (X_L - X_C)^2}$

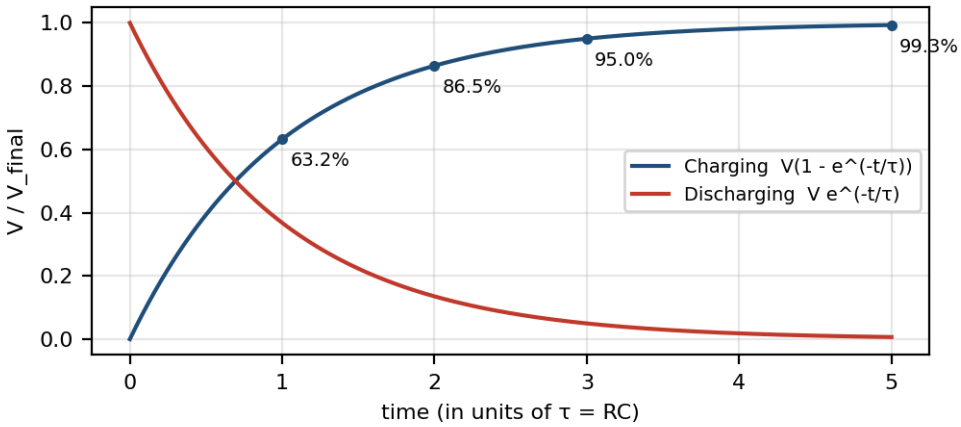
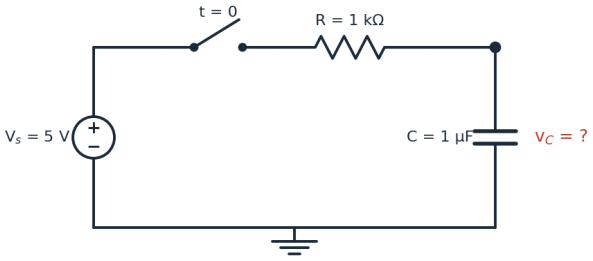


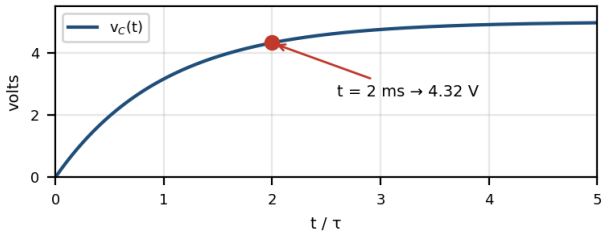
Fig 1.1 – Charging and discharging of an RC circuit (time in units of τ).

1.4 Solved problems

Problem 1.1 – RC step response

Question: The switch closes at t = 0 and a 5 V source is applied to R = 1 kΩ and C = 1 µF (initially uncharged). Find (a) the time constant, (b) v_C at t = 2 ms, (c) the time to reach 90 % of 5 V.





Solution

Step 1. Time constant: $\tau = RC = 1000 \times 1{\times}10^{-6} = 1{\times}10^{-3}$ s = **1 ms**.

Step 2. Capacitor voltage: $v_C(t) = V_s(1 - e^{-t/\tau})$.

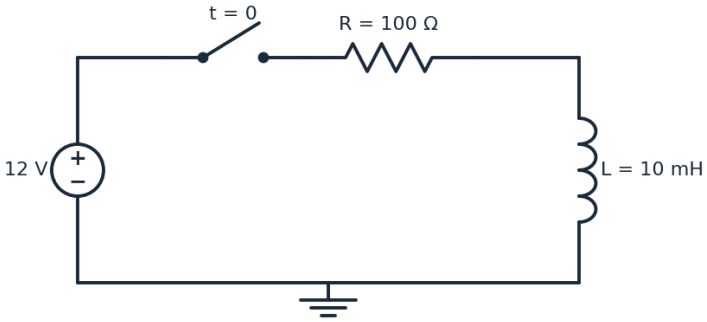
Step 3. At t = 2 ms = 2τ: $v_C = 5(1 - e^{-2}) = 5(1 - 0.1353) =$ **4.32 V**.

Step 4. For 90 %: $t = \tau \cdot \ln(10) = 2.303 \times 1$ ms = **2.3 ms**.

Answer: τ = 1 ms; $v_C(2\text{ ms}) = 4.32$ V; $t_{90\,\%} = 2.3$ ms.

Problem 1.2 – RL circuit – current build-up

Question: A 12 V source is switched onto R = 100 Ω in series with L = 10 mH. Find the time constant, the final current, the current after one time constant, and the inductor voltage at $t = 0^+$ and $t = \infty$.



Solution

Step 1. $\tau = L/R = 10\ \text{mH} / 100\ \Omega =$ **100 µs**.

Step 2. At $t = \infty$ the inductor is a short circuit, so $I_{final} = V/R = 12/100 =$ **120 mA**.

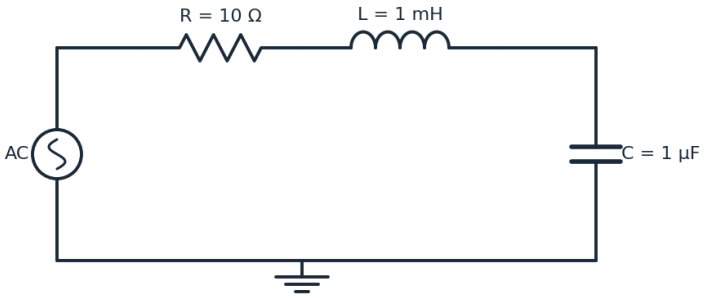
Step 3. At $t = \tau$: $i = 0.632 \times 120\ \text{mA} =$ **75.8 mA** (at 3τ it is 95 % = 114 mA).

Step 4. At $t = 0^+$ the current is still 0, so no voltage drops across R and the whole 12 V appears across L ($v_L = 12$ V). At $t = \infty$ $di/dt = 0$, so $v_L = 0$ V.

Answer: $\tau = 100$ µs; $I_{final} = 120$ mA; $i(\tau) = 75.8$ mA; $v_L = 12\ \text{V} \rightarrow 0$ V.

Problem 1.3 – Series RLC – resonance

Question: An AC source drives R = 10 Ω, L = 1 mH and C = 1 µF in series. Find the resonant frequency, X_L and X_C at resonance, the impedance, Q and bandwidth.



Solution

Step 1. $f_0 = 1/(2\pi\sqrt{(LC)}) = 1/(2\pi\sqrt{(10^{-3} \times 10^{-6})}) = 1/(2\pi \times 3.162\times10^{-5}) =$ **5.03 kHz**.

Step 2. $X_L = 2\pi f_0 L = 2\pi \times 5033 \times 10^{-3} =$ **31.6 Ω**; $X_C = 1/(2\pi f_0 C) =$ **31.6 Ω** – equal, so they cancel.

Step 3. At resonance $|Z| = R =$ **10 Ω** (minimum), so the current is maximum = V/R.

Step 4. $Q = X_L/R = 31.6/10 =$ **3.16**; $BW = f_0/Q = 5033/3.16 =$ **1.59 kHz**.

Answer: $f_0 = 5.03$ kHz; $X_L = X_C = 31.6\ \Omega$; $Z = 10\ \Omega$; $Q = 3.16$; BW = 1.59 kHz.

Problem 1.4 – Capacitors in series – voltage sharing

Question: A 10 V DC source is connected across points A–B. Find the equivalent capacitance and the voltage across each capacitor.



Solution

Step 1. $C_{eq} = (C_1C_2)/(C_1 + C_2) = (10 \times 40)/50 =$ **8 µF**.

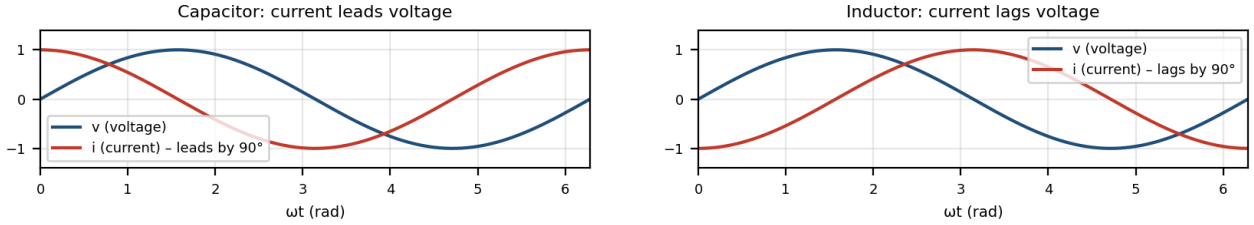
Step 2. Series capacitors carry the *same charge*: $Q = C_{eq}V = 8\ \mu\text{F} \times 10\ \text{V} = 80\ \mu\text{C}$.

Step 3. $V_1 = Q/C_1 = 80/10 =$ **8 V**; $V_2 = Q/C_2 = 80/40 =$ **2 V** (sum = 10 V ✓).

Answer: $C_{eq} = 8$ µF; the smaller capacitor takes the larger voltage (8 V and 2 V).

Problem 1.5 – Reactance and phase of C and L

Question: A 10 V peak, 1 kHz sine wave is applied (separately) across a 1 µF capacitor and across a 10 mH inductor. Find the peak current in each and its phase relative to the voltage.



Solution

Step 1. Capacitor: $X_C = 1/(2\pi \times 1000 \times 10^{-6}) = 159.2\ \Omega \rightarrow I_{pk} = 10/159.2 =$ **62.8 mA**; current **leads** by 90°.

Step 2. Inductor: $X_L = 2\pi \times 1000 \times 0.01 = 62.8\ \Omega \rightarrow I_{pk} = 10/62.8 =$ **159 mA**; current **lags** by 90°.

Step 3. Frequency dependence: if f doubles, X_C halves (current rises) while X_L doubles (current falls).

Answer: C: 62.8 mA, leading 90°; L: 159 mA, lagging 90°.

Quick concept questions

Q. Why does a capacitor block DC but pass AC?

A. In steady-state DC no current flows ($i = C \cdot dv/dt = 0$ when v is constant). For AC the voltage keeps changing, so current keeps flowing; X_C falls as frequency rises.

Q. Can the voltage across a capacitor jump instantly?

A. No – it would need infinite current. The same is true of the current through an inductor.

2. Network and Circuit Theory

In one line: network theory gives systematic ways to find voltages and currents in circuits made of sources and passive components, using Ohm's law, Kirchhoff's laws and a few powerful theorems.

2.1 Core laws

- **Ohm's law:** $V = IR$.
- **KCL (current law):** the sum of currents entering any node is zero (what flows in flows out).
- **KVL (voltage law):** the sum of voltage rises and drops around any closed loop is zero.

2.2 Types of analysis methods and when to use them

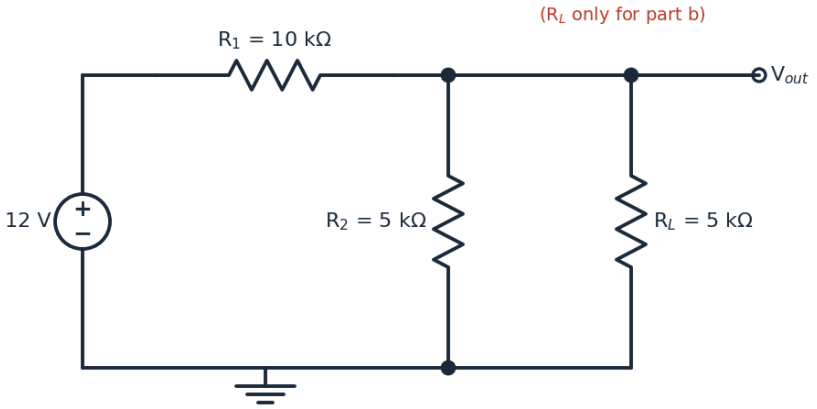
Method	Idea	Use when
Series / parallel reduction	Replace groups of resistors by one equivalent	Simple networks with one source
Voltage / current divider	Voltage splits $\propto R$ in series; current splits $\propto 1/R$ in parallel	Quick check of one node or branch
Nodal analysis (KCL)	Unknowns = node voltages	Many parallel branches / current sources
Mesh analysis (KVL)	Unknowns = loop currents	Many series loops / voltage sources
Superposition	One independent source at a time (others: V → short, I → open); add results	Several sources, linear circuit
Thevenin / Norton	Reduce any network to $V_{th} + R_{th}$ (or $I_N \parallel R_{th}$) as seen by a load	Changing load, max power, source modelling
Max power transfer	$R_L = R_{th}$ gives $P_{max} = V_{th}^2/4R_{th}$ (efficiency 50 %)	Matching source to load (RF, audio)
Delta–Wye	$R_Y = R_\Delta/3$ for equal resistors	Bridge circuits

Key formulae	
Series / parallel R	$R_s = R_1 + R_2\ldots$; $1/R_p = 1/R_1 + 1/R_2\ldots$; two resistors: $R_1R_2/(R_1+R_2)$
Voltage divider	$V_{out} = V_{in}\cdot R_2/(R_1 + R_2)$ (across R_2); with load: use $R_2 \parallel R_L$
Current divider	$I_1 = I\cdot R_2/(R_1 + R_2)$ (the *other* resistor on top)
Power	$P = VI = I^2R = V^2/R$
Max power	$R_L = R_{th}$, $P_{max} = V_{th}^2/(4R_{th})$
LED resistor	$R = (V_{supply} - V_F)/I_{LED}$

2.3 Solved problems

Problem 2.1 – Voltage divider and the loading effect

Question: V_{in} = 12 V, R_1 = 10 kΩ, R_2 = 5 kΩ. Find V_{out} (a) without load and (b) when a 5 kΩ load R_L is connected.



Solution

Step 1. (a) $V_{out} = 12 \times 5k/(10k + 5k) =$ **4 V**.

Step 2. (b) The load is in parallel with R_2: $R_2 \| R_L = (5k \times 5k)/(10k) = 2.5$ kΩ.

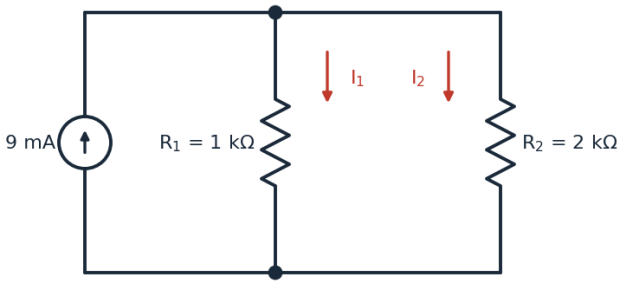
Step 3. $V_{out} = 12 \times 2.5k/(10k + 2.5k) = 12 \times 0.2 =$ **2.4 V**.

Step 4. Lesson: a loaded divider drops. Keep $R_L \gg R_2$ (at least 10×) if the output must stay accurate.

Answer: (a) 4 V; (b) 2.4 V.

Problem 2.2 – Current divider

Question: A 9 mA current source feeds R_1 = 1 kΩ in parallel with R_2 = 2 kΩ. Find I_1, I_2 and the voltage across the pair.



Solution

Step 1. $I_1 = 9$ mA $\times R_2/(R_1 + R_2) = 9 \times 2/3 =$ **6 mA**.

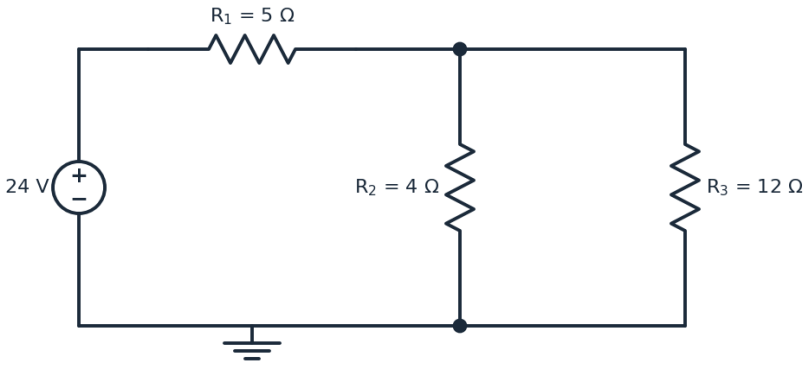
Step 2. $I_2 = 9$ mA $\times R_1/(R_1 + R_2) = 9 \times 1/3 =$ **3 mA**.

Step 3. Check: $I_1 + I_2 = 9$ mA ✓. Voltage $= I_1 R_1 = 6$ mA × 1 kΩ = **6 V** $= I_2 R_2 = 3$ mA × 2 kΩ ✓.

Answer: I_1 = 6 mA, I_2 = 3 mA, V = 6 V (the smaller resistor carries the larger current).

Problem 2.3 – Series–parallel network

Question: Find the total resistance, the source current, and the current and power in each resistor (R_1 = 5 Ω, R_2 = 4 Ω, R_3 = 12 Ω, source 24 V).



Solution

Step 1. $R_2 \| R_3 = (4 \times 12)/(4 + 12) = 3\ \Omega$.

Step 2. $R_{total} = R_1 + 3 = \mathbf{8\ \Omega}$.

Step 3. $I = 24/8 = \mathbf{3\ A}$ (this flows through R_1). $P_{R1} = I^2R = 9 \times 5 = 45$ W.

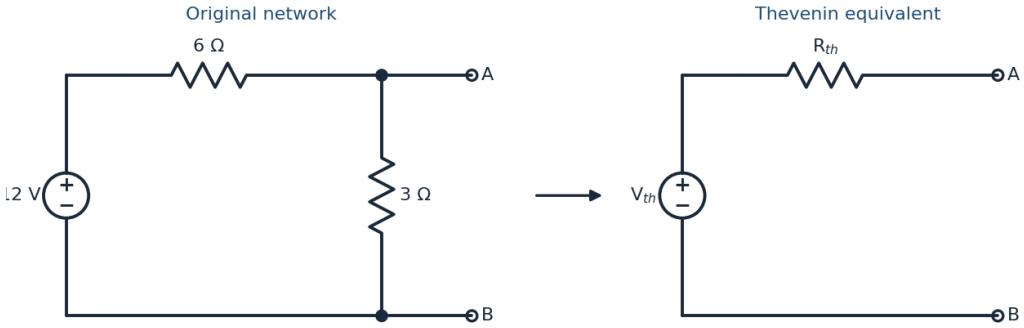
Step 4. Voltage across the parallel pair = 3 A × 3 Ω = 9 V → $I_2 = 9/4 = \mathbf{2.25\ A}$, $I_3 = 9/12 = \mathbf{0.75\ A}$ (sum 3 A ✓).

Step 5. $P_{R2} = 9 \times 2.25 = 20.25$ W; $P_{R3} = 9 \times 0.75 = 6.75$ W. Total = 45 + 20.25 + 6.75 = 72 W = 24 V × 3 A ✓.

Answer: $R_{total} = 8\ \Omega$; $I = 3$ A; $I_2 = 2.25$ A; $I_3 = 0.75$ A; power balance 72 W.

Problem 2.4 – Thevenin equivalent and maximum power

Question: A 12 V source with a 6 Ω series resistor feeds a 3 Ω resistor across terminals A–B. Find the Thevenin equivalent seen by a load at A–B and the maximum power the load can receive.



Solution

Step 1. $\mathbf{V_{th}}$ = open-circuit voltage across A–B = 12 × 3/(6 + 3) = **4 V**.

Step 2. $\mathbf{R_{th}}$: switch the source off (replace by a short) → 6 Ω ∥ 3 Ω = 18/9 = **2 Ω**.

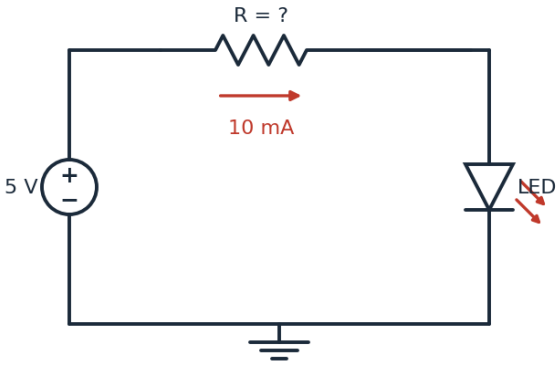
Step 3. Max power transfer when $R_L = R_{th} = 2\ \Omega$: $I = 4/(2 + 2) = 1$ A.

Step 4. $P_{max} = V_{th}^2/(4R_{th}) = 16/8 = \mathbf{2\ W}$ (check: $I^2R_L = 1 \times 2 = 2$ W ✓).

Answer: $V_{th} = 4$ V, $R_{th} = 2\ \Omega$; $P_{max} = 2$ W at $R_L = 2\ \Omega$.

Problem 2.5 – LED current-limiting resistor

Question: Choose a resistor so that a 2 V LED carries 10 mA from a 5 V supply. Use the standard E12 value and check the actual current and the resistor power.



Solution

Step 1. Voltage across R = 5 − 2 = 3 V. R = 3 V/10 mA = **300 Ω**.

Step 2. Nearest standard (E12) value: 330 Ω → I = 3/330 = **9.1 mA** (slightly dimmer, safe).

Step 3. $P_R = I^2R = (9.1\text{ mA})^2 \times 330 =$ **27 mW** – a ¼ W resistor is plenty.

Answer: R = 300 Ω ideal; use 330 Ω (I = 9.1 mA, P = 27 mW).

Quick concept questions

Q. Which is larger – current in a 1 kΩ or a 10 kΩ resistor connected in parallel?

A. The 1 kΩ: same voltage, lower resistance → more current.

Q. In KCL, a node has 5 mA in, 2 mA and 1 mA out through two branches. What is the third branch current?

A. 5 = 2 + 1 + I → I = 2 mA (out).

Q. Delta of three 30 Ω resistors converts to what Wye?

A. $R_Y = 30/3 = 10\ \Omega$ each.

3. Diodes and Their Applications

In one line: a diode is a one-way valve for current: it conducts when forward biased (anode more positive than cathode by about 0.7 V for silicon) and blocks when reverse biased.

3.1 Types of diodes – when to use what

Type	Key property	Use it for
Silicon rectifier (1N400x)	$V_F \approx 0.7$ V; handles amps; slow recovery	Mains rectification, reverse-polarity protection
Schottky	$V_F \approx 0.2$–0.4 V; very fast; more leakage	Low-drop rectifier, DC-DC converters, OR-ing of supplies
Zener	Conducts in reverse at a set voltage V_Z	Voltage reference/regulation, over-voltage clamp
LED	V_F 1.8–3.3 V; emits light	Indicators (always with a series resistor)
Photodiode	Reverse current ∝ light	Light sensors, optical receivers
TVS diode	Very fast avalanche clamp	ESD / surge protection on I/O pins
Fast switching (1N4148)	Recovery in nanoseconds	Signal clipping, logic, snubbers
Varactor	Capacitance changes with reverse bias	Tuners, VCOs

3.2 Types of diode circuits

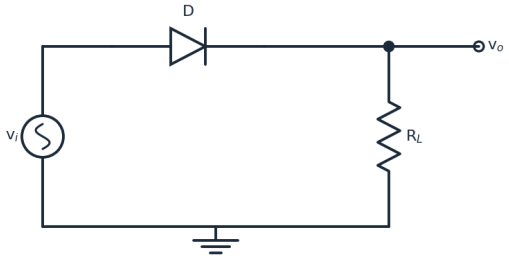
Circuit	What it does	Application
Rectifier (half-wave, full-wave centre-tap, bridge)	AC → pulsating DC	Power supplies
Filter capacitor	Smooths the pulses; ripple $V_r \approx I/(fC)$	After the rectifier
Clipper (limiter)	Cuts off part of the waveform above/below a level; shape is cut	Protects inputs from over-voltage, wave shaping
Clamper (DC restorer)	Shifts the whole waveform up/down; shape unchanged	Level shifting, video DC restore
Voltage multiplier	Capacitor-diode ladder gives 2×, 3× peak	High-voltage from low AC
Protection	Flyback diode across coils; TVS on pins; reverse-polarity diode; input clamps	Relays, motors, I/O pins

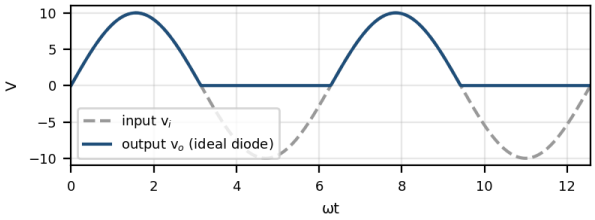
Key formulae	
Shockley equation	$I = I_S(e^{V/(nV_T)} - 1)$; $V_T = kT/q \approx 25.85$ mV (300 K); ≈ 60 mV per decade of current
Half-wave rectifier	$V_{dc} = V_m/\pi = 0.318V_m$; $V_{rms} = V_m/2$; PIV = V_m; ripple factor 1.21; $\eta_{max} = 40.6$ %
Full-wave rectifier	$V_{dc} = 2V_m/\pi = 0.637V_m$; $V_{rms} = V_m/\sqrt{2}$; ripple 0.48; $\eta_{max} = 81.2$ %; PIV: bridge = V_m, centre-tap = $2V_m$
Bridge output peak	$V_{m,out} = V_{m,sec} - 2V_F$
Ripple with filter	$V_{r(pp)} \approx I_{load}/(f_{ripple} \cdot C)$; $f_{ripple} = 100$ Hz (full-wave, 50 Hz mains)
Zener regulator	$R_s = (V_{in} - V_Z)/(I_{Z,min} + I_{L,max})$; $P_Z = V_Z I_Z$
Inductive spike	$v = L \cdot di/dt$ (the reason for a flyback diode)

3.3 Solved problems

Problem 3.1 – Half-wave rectifier

Question: A sine wave of 10 V peak feeds the circuit. Find V_{dc}, V_{rms} of the output and the diode PIV, (a) for an ideal diode and (b) with a 0.7 V silicon diode.





Solution

Step 1. (a) Ideal: $V_{dc} = V_m/\pi = 10/3.1416 =$ **3.18 V**; $V_{rms} = V_m/2 =$ **5 V**.

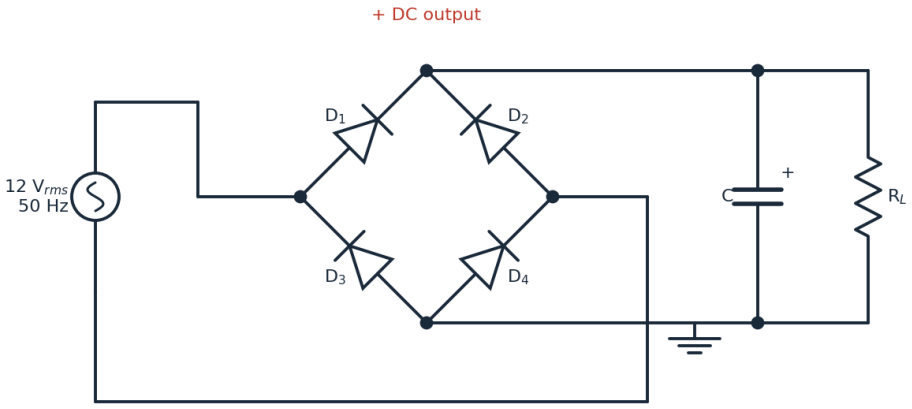
Step 2. (b) Real diode: the output peak drops to $10 - 0.7 = 9.3$ V → $V_{dc} = 9.3/\pi =$ **2.96 V**.

Step 3. PIV: when the input is at its negative peak the diode must block the full $-V_m$, so PIV = **10 V** (choose a diode rated ≥ 1.5×).

Answer: Ideal: 3.18 V dc, 5 V rms; with 0.7 V drop: 2.96 V dc; PIV = 10 V.

Problem 3.2 – Bridge rectifier with filter capacitor

Question: A 12 V_{rms}, 50 Hz transformer secondary feeds a bridge rectifier (silicon diodes) with a 2200 µF filter capacitor and a 500 mA load. Find the peak DC output voltage, the ripple frequency and the peak-to-peak ripple.



Solution

Step 1. Secondary peak: $V_m = 12 \times \sqrt{2} = 16.97$ V.

Step 2. Two diodes conduct at any time, so output peak = $16.97 - 2 \times 0.7 =$ **15.57 V**.

Step 3. Full-wave rectifies both half-cycles: ripple frequency = $2 \times 50 =$ **100 Hz**.

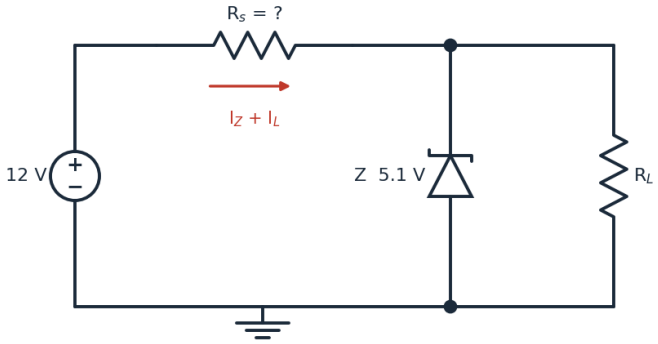
Step 4. $V_{r(pp)} = I/(fC) = 0.5/(100 \times 2200\times10^{-6}) = 0.5/0.22 =$ **2.27 V**.

Step 5. Average DC ≈ $15.57 - 2.27/2 = 14.4$ V. To halve the ripple, double C (4400 µF).

Answer: Peak 15.57 V; ripple 100 Hz; $V_r = 2.27$ V p-p; $V_{dc} \approx 14.4$ V.

Problem 3.3 – Zener shunt regulator design

Question: Design a 5.1 V regulator from a 12 V supply for a load of up to 20 mA. The Zener needs at least 5 mA to stay in regulation. Find R_s (use the next standard value) and the worst-case Zener power.



Solution

Step 1. Minimum Zener current + maximum load current: 5 + 20 = 25 mA must flow through R_s.

Step 2. R_s = (12 − 5.1)/25 mA = 6.9/0.025 = 276 Ω → choose **270 Ω** (a lower value guarantees ≥ 25 mA).

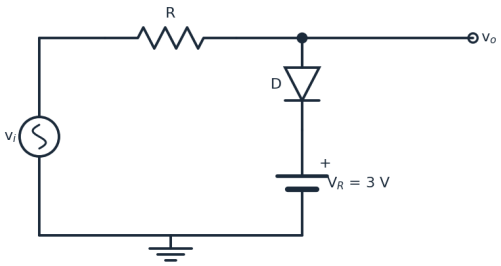
Step 3. Worst case = load disconnected, all current goes through the Zener: I = 6.9/270 = 25.6 mA.

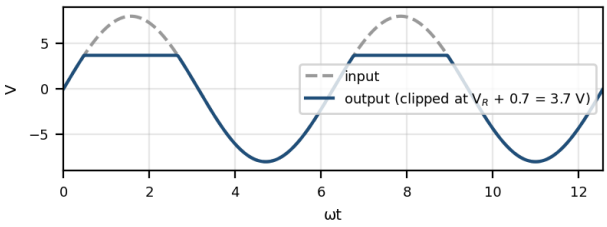
Step 4. P_Z = 5.1 V × 25.6 mA = **130 mW** → use a 500 mW Zener. Resistor power = $(6.9)^2/270$ = 0.18 W.

Answer: R_s = 270 Ω; Zener dissipates 130 mW at no load.

Problem 3.4 – Clipper

Question: An 8 V peak sine wave drives the clipper. The reference battery is 3 V and the diode drops 0.7 V. Sketch/describe the output.





Solution

Step 1. The diode conducts when v_o exceeds V_R + 0.7 = 3.7 V.

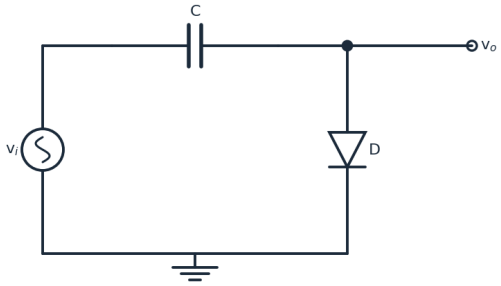
Step 2. While conducting, the output is held at **3.7 V** (positive peaks flattened). The resistor R drops the rest.

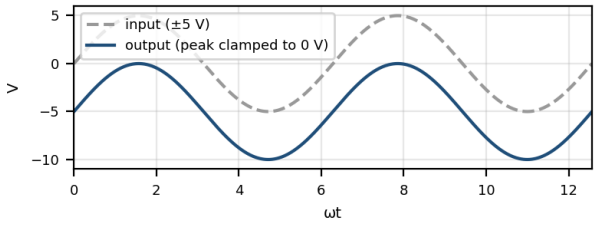
Step 3. When v_i < 3.7 V the diode is off and the output follows the input, including the whole negative half (down to −8 V).

Answer: Output = input, clipped at +3.7 V; negative half unchanged (−8 V peak).

Problem 3.5 – Clamper

Question: A 5 V peak sine wave is applied to the capacitor-diode circuit (ideal diode, large C). Find the output waveform's DC level and its minimum and maximum.





Solution

Step 1. On the first positive peak the diode conducts and charges C to **+5 V** (left plate positive relative to the right).

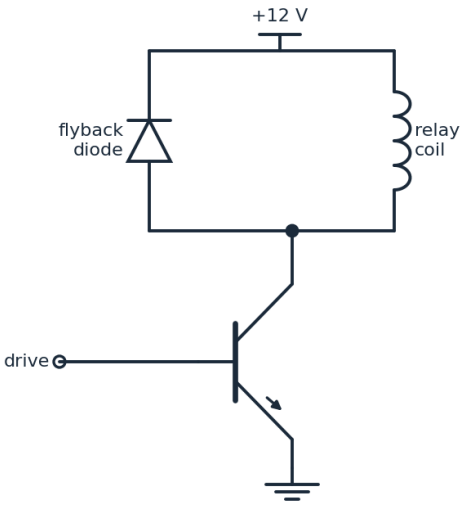
Step 2. Afterwards the capacitor acts as a 5 V battery in series with the source: $v_o = v_i - 5$.

Step 3. So the waveform keeps its shape (10 V p-p) but is shifted down: maximum = 0 V, minimum = **−10 V**, DC level = −5 V.

Answer: Output swings between −10 V and 0 V (positive peaks clamped to 0 V).

Problem 3.6 – Flyback (freewheel) diode

Question: A relay coil (L = 100 mH, R_{coil} = 240 Ω) is driven from 12 V by a transistor. Find the coil current and explain what happens at turn-off with and without the flyback diode (assume the switch opens in 1 µs).



Solution

Step 1. Steady current: I = 12/240 = **50 mA**.

Step 2. At turn-off the inductor current cannot stop instantly: v = L·di/dt. Without a diode: v = 0.1 H × 0.05 A/1 µs = **5000 V** – enough to destroy the transistor.

Step 3. With the diode, the coil current circulates through the diode, so the collector rises only to 12 V + 0.7 V = **12.7 V** and the current decays with τ = L/R = 0.1/240 = 0.42 ms.

Answer: 50 mA coil current; spike ≈ 5 kV without the diode, clamped to 12.7 V with it.

Quick concept questions

Q. By what factor does a diode current change if V rises by 60 mV (n = 1, 300 K)?

A. $e^{60/25.85} = e^{2.32} \approx$ **10×** (one decade per ≈ 60 mV).

Q. Difference between a clipper and a clamper?

A. A clipper removes part of the waveform; a clamper moves the entire waveform up or down without changing its shape.

Q. Why choose a Schottky diode in a switching converter?

A. Lower forward drop (less loss) and almost no reverse-recovery time.

4. Transistors – BJT, FET and CMOS

In one line: a transistor is a controlled valve – a small base current (BJT) or gate voltage (FET) controls a larger current. It works as an **amplifier** (active region) or a **switch** (cut-off/saturation).

4.1 Types of transistors

Type	Control / polarity	Notes and uses
BJT – npn	Current-controlled; ON when $V_{BE} \approx +0.7$ V; current flows C → E	Most common; amplifiers, switches (low-side)
BJT – pnp	ON when $V_{BE} \approx -0.7$ V; current flows E → C	High-side switches, complementary amplifier pairs
JFET (n/p channel)	Voltage-controlled; normally ON (depletion); gate junction reverse biased	Low-noise front ends, analog switches
MOSFET – enhancement n/p	Voltage-controlled; nMOS ON for $V_{GS} > V_{th}$; pMOS ON for $V_{GS} < -\lvert V_{th}\rvert$	Digital ICs (CMOS), power switches, no DC gate current
MOSFET – depletion	Conducts at $V_{GS} = 0$	Special analog uses
IGBT	MOS gate + bipolar output	High-voltage, high-power drives

4.2 BJT operating regions and configurations

Region	Condition	Behaviour
Cut-off	$V_{BE} <$ ≈ 0.6 V	No collector current – switch OFF
Active	BE forward, BC reverse; $V_{CE} > 0.2$ V	$I_C = \beta I_B$ – amplifier
Saturation	Both junctions forward; $V_{CE(sat)} \approx 0.2$ V	Fully ON – switch closed

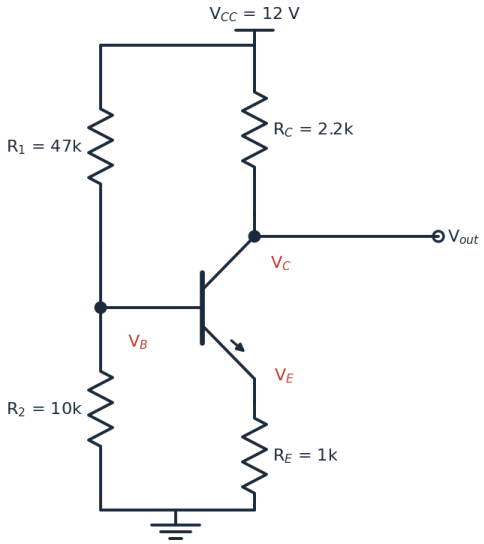
	Common Emitter (CE)	Common Base (CB)	Common Collector (CC)
Input / output	Base / Collector	Emitter / Collector	Base / Emitter
Voltage gain	High, **inverting** (180°)	High, non-inverting	≈ 1, non-inverting (emitter follower)
Current gain	β	α ≈ 1	β + 1
Z_{in} / Z_{out}	Medium / medium-high	**Low** / high	**High** / **low**
Advantage and use	Best power gain; general amplifier	Best high-frequency response; RF, current buffer	Buffer, impedance matching, drives low loads

Key formulae	
BJT currents	$I_E = I_B + I_C$; $I_C = \beta I_B$; $I_E = (\beta+1)I_B$; $\alpha = \beta/(\beta+1)$; $\beta = \alpha/(1-\alpha)$
Small signal	$g_m = I_C/V_T$; $r_\pi = \beta/g_m$; $r_e = V_T/I_E$
Voltage-divider bias	$V_{th} = V_{CC}R_2/(R_1+R_2)$; $R_{th} = R_1 \lVert R_2$; $I_B = (V_{th} - 0.7)/(R_{th} + (\beta+1)R_E)$
CE gain	$A_v = -g_m(R_C \lVert R_L)$ (emitter bypassed)
Switch design	$I_B \geq I_C/\beta_{min}$ (use 2–5× overdrive); $R_B = (V_{drive} - 0.7)/I_B$
MOSFET saturation	$I_D = \tfrac{1}{2}k(V_{GS} - V_{th})^2$, valid when $V_{DS} \geq V_{ov} = V_{GS} - V_{th}$
MOSFET triode	$I_D = k[(V_{GS} - V_{th})V_{DS} - V_{DS}^2/2]$
MOSFET g_m	$g_m = k \cdot V_{ov} = 2I_D/V_{ov}$
CMOS dynamic power	$P = \alpha \cdot C_L \cdot V_{DD}^2 \cdot f$

4.3 Solved problems

Problem 4.1 – BJT voltage-divider bias (given β)

Question: V_{CC} = 12 V, R_1 = 47 kΩ, R_2 = 10 kΩ, R_C = 2.2 kΩ, R_E = 1 kΩ, β = 100, V_{BE} = 0.7 V. Find I_B, I_C, V_E, V_C, V_{CE} and confirm the transistor is in the active region.



Solution

Step 1. Thevenin of the base divider: $V_{th} = 12 \times 10/57 =$ **2.105 V**; $R_{th} = 47k \parallel 10k =$ **8.25 kΩ**.

Step 2. Base loop: $V_{th} = I_B R_{th} + 0.7 + I_E R_E$ with $I_E = (\beta+1) I_B$.

Step 3. $I_B = (2.105 - 0.7)/(8.25k + 101 \times 1k) = 1.405/109.25k =$ **12.9 µA**.

Step 4. $I_C = \beta I_B =$ **1.29 mA**; $I_E = 1.30$ mA.

Step 5. $V_E = I_E R_E =$ **1.30 V**; $V_C = 12 - 1.29\text{ mA} \times 2.2k =$ **9.17 V**; $V_{CE} = 9.17 - 1.30 =$ **7.87 V**.

Step 6. $V_{CE} = 7.87\text{ V} \gg 0.2\text{ V}$ and V_{BE} forward → **active region** ✓ (a quick estimate: $V_E \approx V_B - 0.7 = 1.4$ V gives $I_C \approx 1.4$ mA).

Answer: I_C = 1.29 mA, V_{CE} = 7.87 V – properly biased in the middle of the supply range.

Problem 4.2 – CE amplifier gain and input resistance

Question: For the circuit of Problem 4.1 (R_E bypassed by a large capacitor, no load), find g_m, r_π, the voltage gain and the input resistance seen by the signal.

Solution

Step 1. $g_m = I_C/V_T = 1.29\text{ mA}/25.85\text{ mV} =$ **49.9 mS**.

Step 2. $r_\pi = \beta/g_m = 100/0.0499 =$ **2.0 kΩ**.

Step 3. $A_v = -g_m R_C = -0.0499 \times 2200 =$ **−110** (the minus sign = 180° phase inversion).

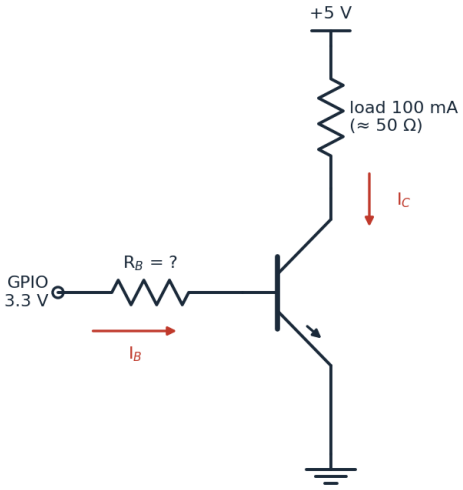
Step 4. $R_{in} = R_1 \parallel R_2 \parallel r_\pi = 8.25k \parallel 2.0k =$ **1.6 kΩ**.

Step 5. If R_E is *not* bypassed: $A_v \approx -R_C/(r_e + R_E) = -2200/(20 + 1000) \approx -2.2$ – gain falls but becomes stable and independent of β.

Answer: g_m = 50 mS, r_π = 2 kΩ, $A_v \approx -110$, $R_{in} \approx 1.6$ kΩ.

Problem 4.3 – BJT as a switch

Question: A 3.3 V GPIO must switch a 100 mA load (5 V supply, ≈ 50 Ω) using an npn BJT with $\beta_{min} = 50$. Choose R_B.



Solution

Step 1. Required base current for saturation: $I_{B,min} = I_C/\beta_{min} = 100\ mA/50 = \mathbf{2\ mA}$.

Step 2. Use overdrive of about 2.5×: $I_B = 5\ mA$, so the switch is surely saturated ($V_{CE(sat)} \approx 0.2\ V$).

Step 3. $R_B = (3.3 - 0.7)/5\ mA = \mathbf{520\ \Omega}$ → standard **510 Ω**.

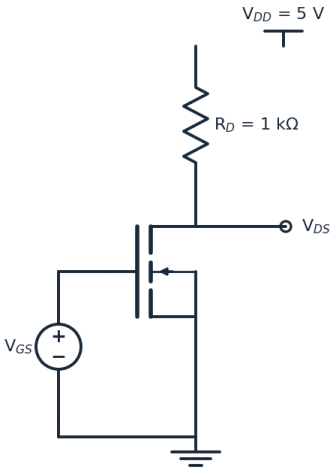
Step 4. Transistor dissipation: $V_{CE(sat)} \times I_C = 0.2 \times 0.1 = 20\ mW$ – negligible.

Step 5. Check that the GPIO can supply 5 mA; if not, use a MOSFET (no DC gate current) or a Darlington. Add a flyback diode if the load is a coil.

Answer: $R_B = 510\ \Omega$ ($I_B \approx 5\ mA$).

Problem 4.4 – NMOS biasing – check the region

Question: An NMOS has $k = \mu_n C_{ox} W/L = 2\ mA/V^2$ and $V_{th} = 1\ V$. With $V_{GS} = 3\ V$, $V_{DD} = 5\ V$ and $R_D = 1\ k\Omega$ find I_D and V_{DS}.



Solution

Step 1. $V_{ov} = V_{GS} - V_{th} = 2\ V$. **Assume saturation:** $I_D = ½ \times 2\ mA/V^2 \times 2^2 = 4\ mA \rightarrow V_{DS} = 5 - 4\ mA \times 1k = 1\ V$.

Step 2. Check: saturation needs $V_{DS} \geq V_{ov} = 2\ V$, but $V_{DS} = 1\ V$ → **assumption wrong; the device is in the triode region**.

Step 3. Triode: $I_D = k[V_{ov}V_{DS} - V_{DS}^2/2]$ with $I_D = (5 - V_{DS})/1k$.

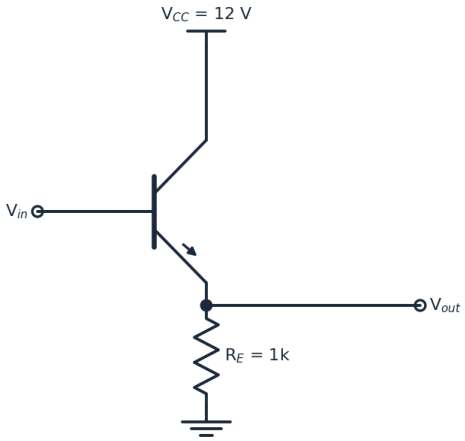
Step 4. $5 - V_{DS} = 2[2V_{DS} - V_{DS}^2/2] = 4V_{DS} - V_{DS}^2 \rightarrow V_{DS}^2 - 5V_{DS} + 5 = 0 \rightarrow V_{DS} = (5 \pm \sqrt{5})/2 = 1.38$ V or 3.62 V.

Step 5. Triode requires $V_{DS} < 2$ V → $\mathbf{V_{DS} = 1.38\ V}$, $I_D = (5 - 1.38)/1k = \mathbf{3.62\ mA}$.

Answer: Triode region: $V_{DS} = 1.38$ V, $I_D = 3.62$ mA. Always verify the region you assumed.

Problem 4.5 – Emitter follower (common collector)

Question: $V_{CC} = 12$ V, $R_E = 1$ kΩ, β = 100. The input DC level is 6.7 V. Find V_E, I_E, the voltage gain and the approximate input resistance.



Solution

Step 1. $V_E = V_B - 0.7 = 6.7 - 0.7 = \mathbf{6.0\ V}$; $I_E = 6$ V/1 kΩ = **6 mA**.

Step 2. $r_e = V_T/I_E = 25.85$ mV/6 mA = 4.3 Ω.

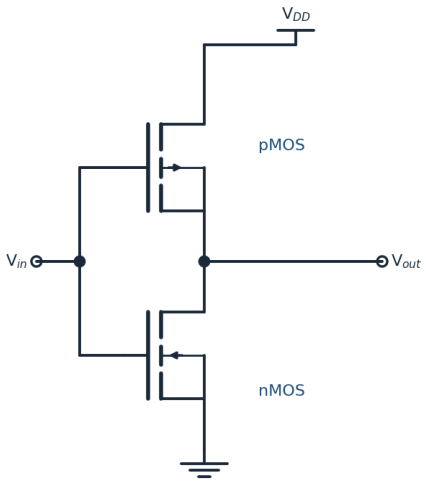
Step 3. $A_v = R_E/(R_E + r_e) = 1000/1004.3 = \mathbf{0.996 \approx 1}$ – output follows the input, no inversion.

Step 4. $R_{in} \approx \beta(r_e + R_E) = 100 \times 1004 \approx \mathbf{100\ k\Omega}$ (high), while output resistance is only a few ohms → an excellent buffer.

Answer: $V_E = 6$ V, $I_E = 6$ mA, $A_v \approx 1$, $R_{in} \approx 100$ kΩ.

Problem 4.6 – CMOS inverter – operation and power

Question: Explain how the CMOS inverter works for $V_{in} = 0$ and $V_{in} = V_{DD}$. If it drives $C_L = 10$ pF at 100 MHz with $V_{DD} = 1.8$ V and activity factor 0.5, find the dynamic power.



Solution

Step 1. V_{in} = 0 (LOW): the pMOS ($V_{GS} = -V_{DD}$) is ON and the nMOS is OFF → output pulled up to $\mathbf{V_{DD}}$.

Step 2. $V_{in} = V_{DD}$ (HIGH): pMOS OFF, nMOS ON → output pulled down to **0 V**. Output is the inverse of the input.

Step 3. In both steady states one transistor is off, so there is no DC path from V_{DD} to ground → **almost zero static power**. Power is used mainly while switching and charging C_L.

Step 4. $P_{dyn} = \alpha\, C_L\, V_{DD}^2\, f = 0.5 \times 10{\times}10^{-12} \times (1.8)^2 \times 10^8 =$ **1.62 mW**.

Step 5. Lowering V_{DD} is the most effective way to cut power because it enters as a square.

Answer: P_{dyn} = 1.62 mW; static power ≈ 0 (leakage only).

Quick concept questions

Q. Give α if β = 99.

A. $\alpha = \beta/(\beta+1) = 0.99$.

Q. Why is an emitter follower useful?

A. Gain ≈ 1, high input and low output resistance – it buffers a weak source so it can drive a heavy load.

Q. MOSFET vs BJT in one sentence each.

A. MOSFET: voltage-controlled, no gate current, dominant in digital/low-power. BJT: current-controlled, higher g_m per current, good analog precision.

5. Op-amps and Voltage Regulators

In one line: an op-amp is a very-high-gain differential amplifier. With **negative feedback** its gain depends only on the external resistors, which makes it the universal building block of analog circuits.

5.1 The two golden rules (ideal op-amp, negative feedback)

- **Rule 1:** no current flows into the input pins (infinite input impedance).
- **Rule 2:** the output adjusts until $V_+ = V_-$ (the "virtual short"). A grounded + input therefore makes the − input a "virtual ground".
- Ideal: infinite gain, infinite bandwidth, zero output impedance, infinite CMRR. Real: finite GBW, slew rate, offset voltage, bias current.

5.2 Types of op-amp circuits

Circuit	Gain / function	Typical use
Inverting amplifier	$V_o = -(R_f/R_{in})V_{in}$	Precise gain, summing node
Non-inverting amplifier	$V_o = (1 + R_f/R_g)V_{in}$	High input impedance gain stage
Voltage follower (buffer)	Gain = 1	Isolate a high-impedance source
Summing amplifier	$V_o = -R_f \Sigma V_i/R_i$	Mixers, DAC
Difference amplifier	$V_o = (R_2/R_1)(V_2 - V_1)$	Remove common-mode noise, sensor bridges
Instrumentation amplifier	Two buffers + difference stage; gain set by one resistor	Small sensor signals, high CMRR
Integrator	$V_o = -(1/RC)\int V_{in}dt$	Ramp generation, filters, PID
Differentiator	$V_o = -RC \cdot dV_{in}/dt$	Edge detection (noisy!)
Comparator (no / positive feedback)	Output saturates to $\pm V_{sat}$	Threshold detection
Schmitt trigger (positive feedback)	Two thresholds → hysteresis	Noise-immune switching
Oscillators (positive feedback)	Loop gain 1 and phase 0° (Barkhausen)	Wien bridge, relaxation

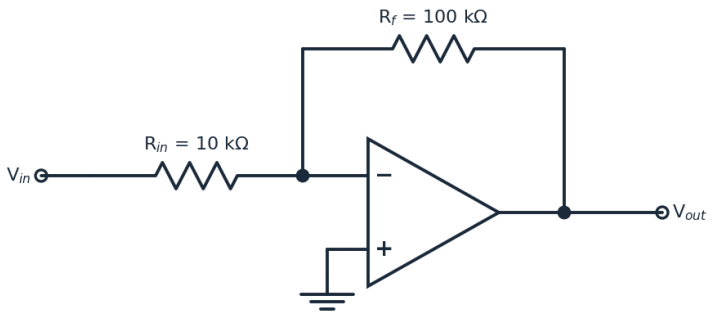
Negative feedback gives stable, predictable gain, wider bandwidth, lower distortion and controlled impedances. **Positive feedback** gives hysteresis, latching and oscillation.

Key formulae	
Closed-loop gain	$A_f = A/(1 + A\beta) \approx 1/\beta$ for large loop gain
Gain-bandwidth	GBW = gain × bandwidth = constant (unity-gain frequency)
Slew rate	$SR = \max dV_o/dt$; $f_{max} = SR/(2\pi V_p)$
CMRR	$CMRR(dB) = 20 \log_{10}(A_d/A_{cm})$
Inverting Schmitt	$V_{TH} = \pm V_{sat} R_1/(R_1 + R_2)$

5.3 Solved problems – op-amp circuits

Problem 5.1 – Inverting amplifier

Question: R_{in} = 10 kΩ, R_f = 100 kΩ, V_{in} = 0.2 V. Find the gain, V_{out} and the input resistance of the stage.



Solution

Step 1. The + input is grounded, so by the virtual short the − input is at 0 V.

Step 2. Current through R_{in}: $I = V_{in}/R_{in} = 0.2/10k = 20\ \mu A$. No current enters the op-amp, so the same 20 µA flows through R_f.

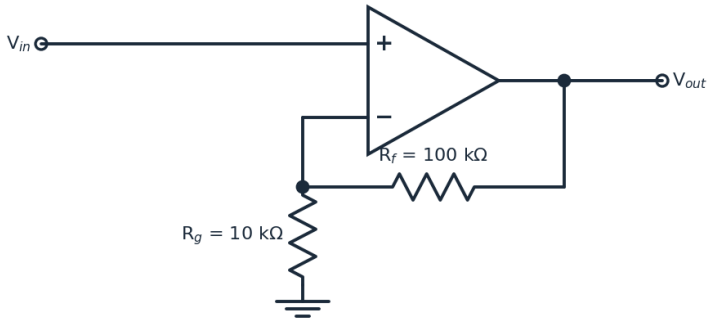
Step 3. $V_{out} = 0 - I \cdot R_f = -20\ \mu A \times 100k = \mathbf{-2\ V}$. Gain $= -R_f/R_{in} = \mathbf{-10}$.

Step 4. Input resistance seen by the source $= R_{in} = \mathbf{10\ k\Omega}$ (not infinite!).

Answer: Gain −10, V_{out} = −2 V, R_{in} = 10 kΩ.

Problem 5.2 – Non-inverting amplifier

Question: R_f = 100 kΩ, R_g = 10 kΩ, V_{in} = 0.5 V. Find the gain and V_{out}. What gain results if R_f is replaced by a short and R_g by an open circuit?



Solution

Step 1. Virtual short: $V_- = V_{in} = 0.5$ V. This is the voltage divider $R_g/(R_g + R_f)$ of V_{out}.

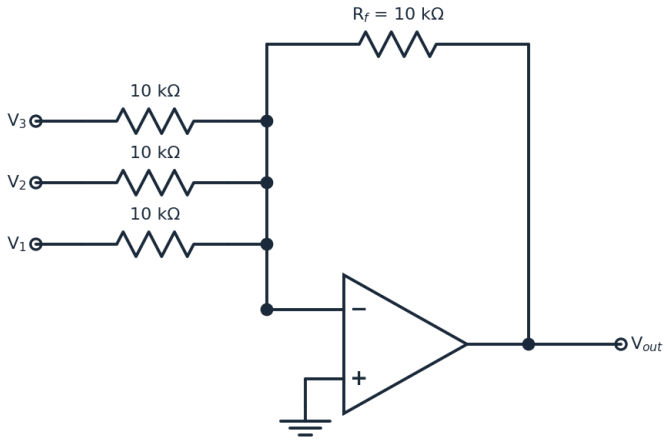
Step 2. $V_{in} = V_{out} \times R_g/(R_g + R_f) \rightarrow V_{out} = V_{in}(1 + R_f/R_g) = 0.5 \times 11 = \mathbf{5.5\ V}$; gain = **11**.

Step 3. $R_f = 0$, $R_g = \infty \rightarrow$ gain = 1: a **voltage follower (buffer)**.

Answer: Gain 11, V_{out} = 5.5 V; the buffer has gain 1.

Problem 5.3 – Summing amplifier

Question: All resistors are 10 kΩ. Inputs are $V_1 = 1$ V, $V_2 = 2$ V, $V_3 = 3$ V. Find V_{out}. What if R_f were 20 kΩ?



Solution

Step 1. The − node is a virtual ground, so each input current is independent: $I_k = V_k/10k = 0.1, 0.2, 0.3$ mA.

Step 2. They add at the node: $I_{total} = 0.6$ mA, which all flows through R_f.

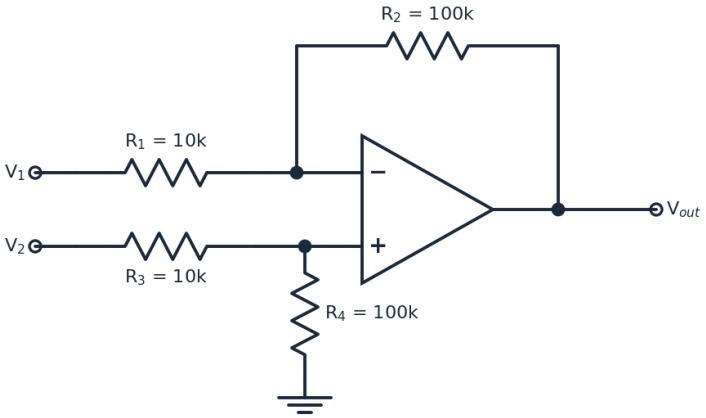
Step 3. $V_{out} = -I_{total}R_f = -0.6$ mA × 10k = **−6 V**.

Step 4. With $R_f = 20$ kΩ: $V_{out} = -12$ V (each input weighted ×2).

Answer: $V_{out} = -6$ V (-12 V with $R_f = 20$ kΩ).

Problem 5.4 – Difference amplifier

Question: $R_1 = R_3 = 10$ kΩ, $R_2 = R_4 = 100$ kΩ, $V_1 = 2.0$ V, $V_2 = 2.3$ V. Find V_{out}.



Solution

Step 1. Matched ratios ($R_2/R_1 = R_4/R_3 = 10$) give $V_{out} = (R_2/R_1)(V_2 - V_1)$.

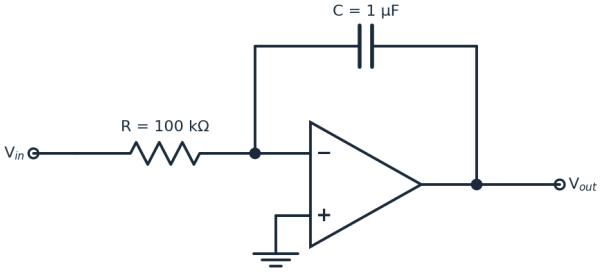
Step 2. $V_{out} = 10 \times (2.3 - 2.0) =$ **3.0 V**. The common-mode part (≈ 2.15 V) is rejected.

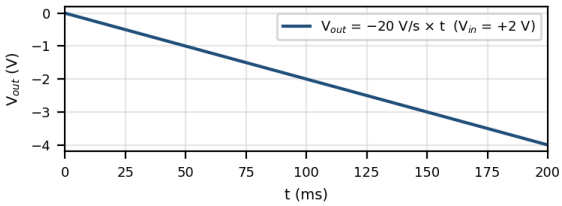
Step 3. Resistor mismatch of only 1 % limits the CMRR to about 40 dB – that is why instrumentation amps use laser-trimmed resistors.

Answer: V_{out} = 3 V (gain 10 on the difference only).

Problem 5.5 – Integrator

Question: R = 100 kΩ, C = 1 µF, V_{in} = +2 V DC applied at t = 0 (capacitor initially uncharged). Find the output slope and the output after 100 ms.





Solution

Step 1. $V_{out}(t) = -(1/RC)\int V_{in}dt$; RC = 100k × 1 µF = 0.1 s.

Step 2. Slope = $-V_{in}/RC = -2/0.1 =$ **−20 V/s** (−20 mV per ms).

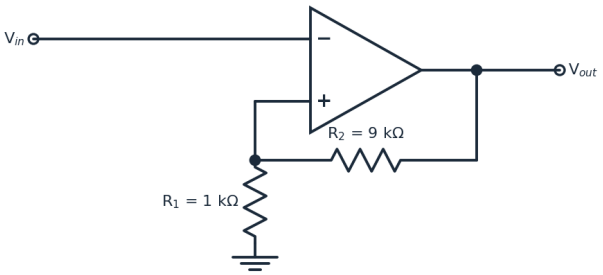
Step 3. After 100 ms: $V_{out} = -20 \times 0.1 =$ **−2 V**. The ramp continues until the op-amp saturates near its supply rail.

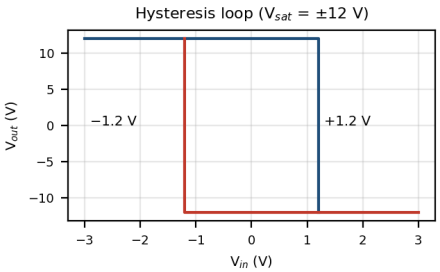
Step 4. Add a large resistor across C in real designs to stop DC offset from saturating the output.

Answer: Output ramps at −20 V/s; −2 V after 100 ms.

Problem 5.6 – Schmitt trigger (positive feedback)

Question: V_{sat} = ±12 V, R_1 = 1 kΩ, R_2 = 9 kΩ. Find the upper and lower threshold voltages and the hysteresis.





Solution

Step 1. The + input is a fraction of the output: $\beta = R_1/(R_1 + R_2) = 1/10 = 0.1$.

Step 2. When V_{out} = +12 V the threshold is $V_{UT} = +12 \times 0.1 =$ **+1.2 V**. When V_{out} = −12 V: $V_{LT} =$ **−1.2 V**.

Step 3. Hysteresis = $V_{UT} - V_{LT} =$ **2.4 V**.

Step 4. Noise smaller than 2.4 V p-p cannot cause false switching – the reason for using a Schmitt trigger on slow or noisy signals.

Answer: Thresholds ±1.2 V, hysteresis 2.4 V.

5.4 Op-amp limits – quick solved calculations

Gain-bandwidth, slew rate, CMRR

Q. GBW = 1 MHz and the closed-loop gain is 50. What is the bandwidth?

A. BW = GBW/gain = 1 MHz/50 = **20 kHz**.

Q. SR = 0.5 V/µs. Maximum undistorted frequency for a 5 V peak sine?

A. $f_{max} = SR/(2\pi V_p) = 0.5\times10^6/(2\pi \times 5)$ = **15.9 kHz**.

Q. $A_d = 1000$ and $A_{cm} = 0.1$. CMRR in dB?

A. $20 \log(1000/0.1) = 20 \log(10^4)$ = **80 dB**.

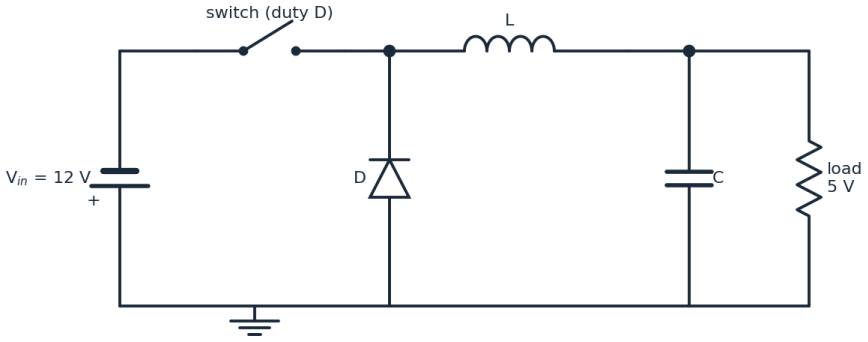
5.5 Types of voltage regulators

Regulator	How it works	Pros / cons
Zener shunt	Zener + series resistor	Simple, wasteful, poor load regulation
Linear series (78xx, LM317, LDO)	Pass transistor controlled by an error amplifier against a reference	Low noise, simple; efficiency = V_{out}/V_{in}; heat = $(V_{in} - V_{out})I$. 78xx needs ≈ 2 V dropout, LDO 0.1–0.3 V
Switching buck (step-down)	Switch + inductor + diode, PWM duty D	85–95 % efficient; ripple/EMI to manage
Switching boost (step-up)	Inductor stores energy then adds to input	Output > input
Buck-boost	Inverting or non-inverting	Output above or below input

Key formulae	
Linear regulator	$P_{diss} = (V_{in} - V_{out})\cdot I_L$; $\eta = V_{out}/V_{in}$
Buck	$V_{out} = D\cdot V_{in}$
Boost	$V_{out} = V_{in}/(1 - D)$
Buck-boost (inverting)	$V_{out} = -V_{in}D/(1 - D)$
LM317	$V_{out} = 1.25\ V \times (1 + R_2/R_1)$
Regulation	Line = $\Delta V_{out}/\Delta V_{in}$; Load = $\Delta V_{out}/\Delta I_{load}$

Problem 5.7 – Buck converter vs linear regulator

Question: Convert 12 V to 5 V at 500 mA. (a) What duty cycle does a buck converter need? (b) Compare the power lost in a linear regulator with a buck converter that is 90 % efficient.



Solution

Step 1. (a) $D = V_{out}/V_{in} = 5/12$ = **0.417** (41.7 %).

Step 2. (b) Output power = 5 V × 0.5 A = 2.5 W.

Step 3. Linear: the excess voltage is burned as heat: P = (12 − 5) × 0.5 = **3.5 W**; efficiency = 5/12 = **41.7 %** (needs a heat sink).

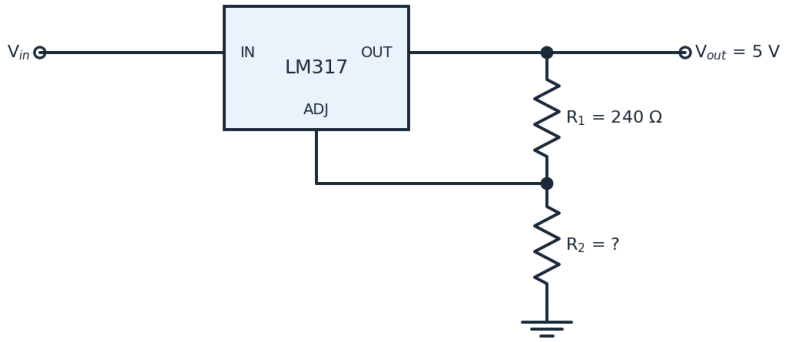
Step 4. Buck: $P_{in} = 2.5/0.9 = 2.78$ W → loss = **0.28 W** – twelve times less heat.

Step 5. How it works: the switch connects the input to the inductor for a fraction D of each cycle; the inductor/capacitor average the pulses into a steady DC voltage; the diode gives the inductor current a path while the switch is open.

Answer: D = 0.417; linear loses 3.5 W (η 41.7 %), buck loses 0.28 W (η 90 %).

Problem 5.8 – Adjustable LM317

Question: Set an LM317 to give 5 V using R_1 = 240 Ω. Find R_2. How much power does the regulator waste at 200 mA if V_{in} = 9 V?



Solution

Step 1. The regulator keeps 1.25 V between OUT and ADJ, so the current through R_1 is 1.25/240 = 5.2 mA, which also flows through R_2.

Step 2. $V_{out} = 1.25(1 + R_2/R_1) \rightarrow 5 = 1.25(1 + R_2/240) \rightarrow R_2/240 = 3 \rightarrow$ **R_2 = 720 Ω**.

Step 3. P_{diss} = (9 − 5) × 0.2 = **0.8 W**; efficiency = 5/9 = 55.6 %.

Answer: R_2 = 720 Ω; 0.8 W lost as heat.

6. Filter Circuits – Passive and Active

In one line: a filter passes some frequencies and attenuates others. The **cut-off frequency f_c** is where the output has fallen to 70.7 % (−3 dB) of the pass-band value.

6.1 Types of filters

Type	Passes	Simple circuit / use
Low-pass (LPF)	$f < f_c$	Series R, shunt C. Removes noise, anti-aliasing, smoothing PWM
High-pass (HPF)	$f > f_c$	Series C, shunt R. Blocks DC, AC coupling
Band-pass (BPF)	$f_L < f < f_H$	LPF + HPF or resonant LC. Radio tuning
Band-stop / notch	All except a band	Removes 50 Hz mains hum
All-pass	All, changes phase only	Phase equalisers

	Passive	Active (op-amp)
Parts	R, L, C only	R, C + op-amp (no inductors)
Gain	≤ 1 (loss)	Can amplify; buffered, load independent
Supply	None; high f and high power OK	Needs supply; limited by op-amp GBW
Common responses	Order 1: −20 dB/dec	Butterworth (flat), Chebyshev (ripple, steeper), Bessel (best step response), Sallen-Key / MFB topologies

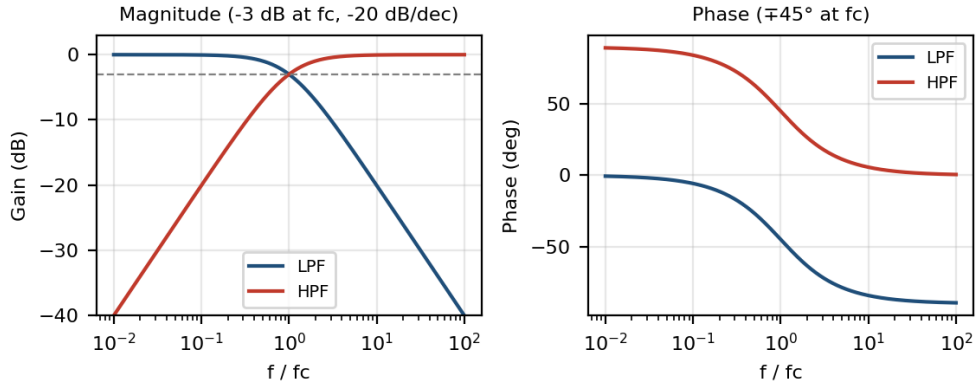


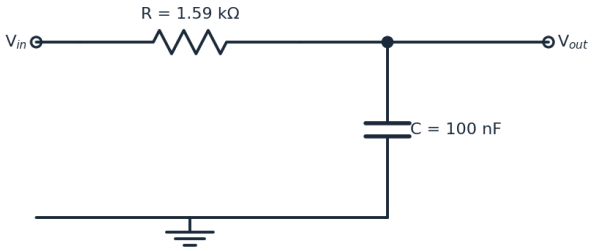
Fig 6.1 – First-order LPF and HPF Bode plots (−3 dB at f_c, ±45° phase).

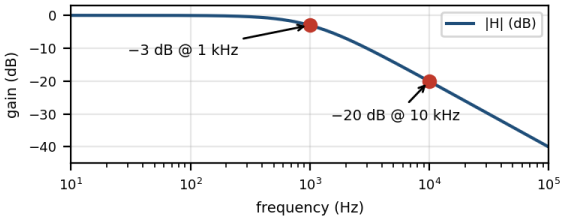
Key formulae	
Cut-off frequency	$f_c = 1/(2\pi RC)$
First-order LPF	$\lvert H\rvert = 1/\sqrt{(1 + (f/f_c)^2)}$; phase = $-\arctan(f/f_c)$
First-order HPF	$\lvert H\rvert = (f/f_c)/\sqrt{(1 + (f/f_c)^2)}$; phase = $+\arctan(f_c/f)$
Roll-off	−20 dB/decade (−6 dB/octave) per pole; n-th order: −20n dB/decade
dB reminder	$dB = 20 \log(V_{out}/V_{in})$; −3 dB = 0.707; −6 dB = 0.5; −20 dB = 0.1
Rise time	$t_r = 2.2RC = 0.35/f_c$
Sallen-Key LPF (unity gain, $R_1 = R_2 = R$)	$f_c = 1/(2\pi R\sqrt{(C_1C_2)})$; $Q = \tfrac{1}{2}\sqrt{(C_1/C_2)}$; Butterworth: Q = 0.707 → $C_1 = 2C_2$
2nd-order Butterworth	$\lvert H\rvert = 1/\sqrt{(1 + (f/f_c)^4)}$; −40 dB/decade
Band-pass	$f_0 = \sqrt{(f_L f_H)}$; $BW = f_H - f_L$; $Q = f_0/BW$

6.2 Solved problems

Problem 6.1 – RC low-pass filter

Question: R = 1.59 kΩ, C = 100 nF. Find the cut-off frequency, the gain (in dB) at 10 kHz, the phase at f_c and the rise time.





Solution

Step 1. $f_c = 1/(2\pi RC) = 1/(2\pi \times 1590 \times 100\times10^{-9}) =$ **1.0 kHz**.

Step 2. At 10 kHz: $f/f_c = 10 \rightarrow |H| = 1/\sqrt{(1 + 100)} = 0.0995 \rightarrow 20 \log(0.0995) =$ **−20 dB** (one decade above f_c, −20 dB per decade ✓).

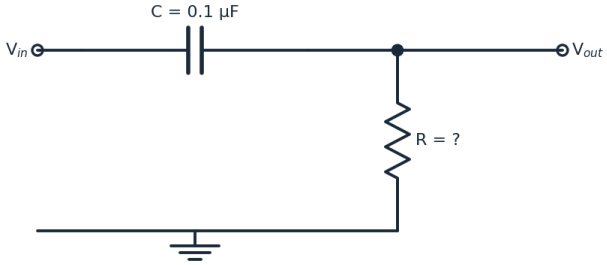
Step 3. Phase at $f_c = -\arctan(1) =$ **−45°**.

Step 4. Rise time $t_r = 2.2RC = 2.2 \times 159$ µs = **350 µs** ($= 0.35/f_c$ ✓).

Answer: f_c = 1 kHz; −20 dB at 10 kHz; −45° at f_c; t_r = 350 µs.

Problem 6.2 – Design an RC high-pass filter

Question: Design a high-pass filter with f_c = 100 Hz using C = 0.1 µF. What is the gain at 10 Hz and the phase at 100 Hz?



Solution

Step 1. $R = 1/(2\pi f_c C) = 1/(2\pi \times 100 \times 0.1\times10^{-6}) =$ **15.9 kΩ** (use 16 kΩ).

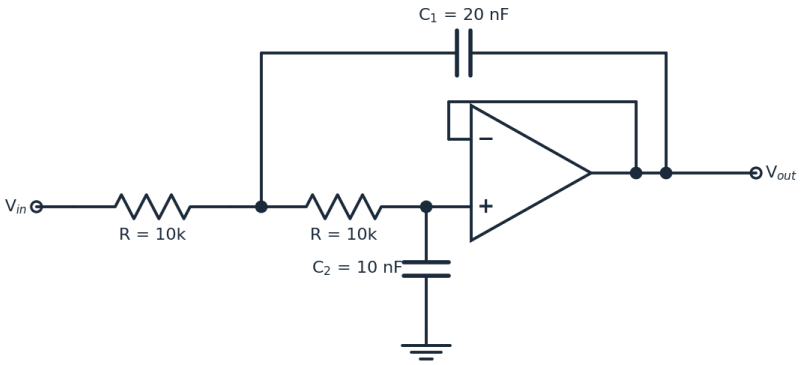
Step 2. At 10 Hz: $f/f_c = 0.1 \rightarrow |H| = 0.1/\sqrt{(1 + 0.01)} = 0.0995 \rightarrow$ **−20 dB** (the signal is strongly attenuated).

Step 3. Phase at $f_c = +\arctan(f_c/f) =$ **+45°** (output leads the input).

Answer: R = 15.9 kΩ; −20 dB at 10 Hz; +45° at 100 Hz.

Problem 6.3 – Sallen-Key Butterworth low-pass

Question: $R_1 = R_2 = 10\ k\Omega$, $C_2 = 10$ nF (to ground) and $C_1 = 20$ nF (to output). Find f_c, Q and the gain at $4\ f_c$.



Solution

Step 1. $f_c = 1/(2\pi R\sqrt{(C_1 C_2)}) = 1/(2\pi \times 10^4 \times \sqrt{(2\times10^{-16})}) = 1/(2\pi \times 10^4 \times 1.414\times10^{-8}) =$ **1.125 kHz**.

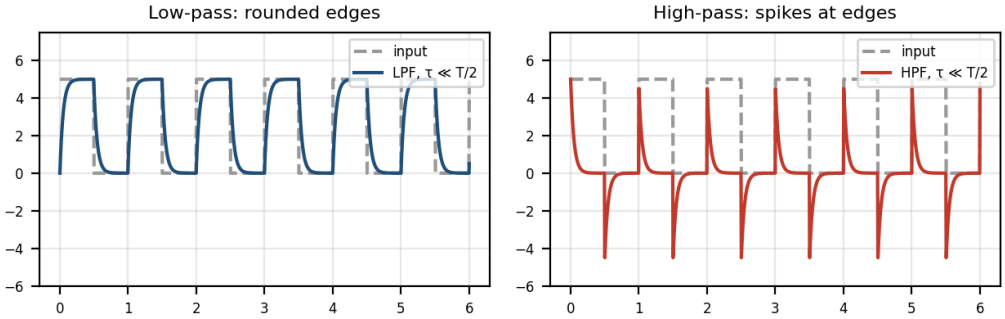
Step 2. $Q = \tfrac{1}{2}\sqrt{(C_1/C_2)} = \tfrac{1}{2}\sqrt{2} =$ **0.707** → maximally flat (Butterworth) response.

Step 3. Second order: $|H| = 1/\sqrt{(1 + (f/f_c)^4)} = 1/\sqrt{(1 + 256)} = 0.0624$ → **−24 dB** at $4\ f_c$ (about −40 dB/decade slope).

Answer: $f_c = 1.125$ kHz, $Q = 0.707$, −24 dB at $4f_c$.

Problem 6.4 – Square-wave response of RC filters (waveform question)

Question: A 5 V square wave with period T is applied to an RC low-pass and to an RC high-pass whose time constant is much smaller than T/2. Describe both outputs. What changes if the time constant is much larger than T?



Solution

Step 1. Low-pass (output across C): each edge makes C charge/discharge exponentially. With $\tau \ll T/2$ the output reaches full amplitude with **rounded edges** (rise time 2.2τ).

Step 2. High-pass (output across R): the capacitor passes only the *change*, so the output is a sharp **spike** (+5 V up, −5 V down) at every edge that decays to 0 with τ – the circuit differentiates.

Step 3. If $\tau \gg T$: the LPF output is only a small triangular ripple around the average (2.5 V) – it **integrates**; the HPF output looks like the input square wave (slight tilt) with its DC removed.

Step 4. Rule: the filter acts as an integrator (LPF) or a differentiator (HPF) when τ is on the wrong side of the signal period.

Answer: LPF: rounded edges / integrator. HPF: spikes at edges / differentiator.

Quick concept questions

Q. Why put a low-pass filter before an ADC?

A. To remove frequencies above $f_s/2$ that would otherwise **alias** into the measured band.

Q. Band-pass with f_L = 900 Hz and f_H = 1.1 kHz. Find f_0, BW and Q.

A. $f_0 = \sqrt{(900 \times 1100)} = 995$ Hz; BW = 200 Hz; Q = 995/200 = **4.97**.

Q. A filter has −3 dB bandwidth 10 kHz. Estimate its rise time.

A. $t_r = 0.35/10$ kHz = **35 µs**.

7. Digital ICs – Logic, Flip-flops, Counters, Registers

In one line: digital ICs work with two voltage levels (0 and 1). **Combinational** circuits depend only on present inputs; **sequential** circuits also remember past inputs using flip-flops and a clock.

7.1 Types of digital ICs

Class	Examples	Notes
Logic gates	AND, OR, NOT, NAND, NOR, XOR, XNOR (74HC00 …)	NAND and NOR are **universal**
Combinational blocks	Mux, demux, decoder, encoder, adder, magnitude comparator	Output = f(inputs) only
Sequential blocks	Latch, flip-flop (SR, D, JK, T), counter, shift register	Edge-triggered storage with clock
Buffers / drivers	74x244, 74x245, ULN2003	Restore levels, isolate, supply more current, tri-state
Comparators	Analog (LM339) vs digital (74x85)	Analog → 1-bit logic; digital compares words
Programmable logic	PROM, PAL, PLA, CPLD, FPGA	PAL: programmable AND; PLA: AND + OR; CPLD: non-volatile; FPGA: SRAM lookup tables
Logic families	TTL (74LS), CMOS (74HC, 74LVC), ECL	CMOS: low static power; compare noise margins and speed

Element	Characteristic equation / behaviour
SR latch	$Q^+ = S + R'Q$ (S = R = 1 forbidden)
D flip-flop	$Q^+ = D$ – copies D at the clock edge; building block of registers
JK flip-flop	$Q^+ = JQ' + K'Q$; J = K = 0 hold, J = K = 1 toggle
T flip-flop	$Q^+ = T \oplus Q$; T = 1 toggles → divide by 2
Latch vs flip-flop	Latch: **level**-sensitive (transparent while enable is high). Flip-flop: **edge**-triggered (samples only at the clock edge)

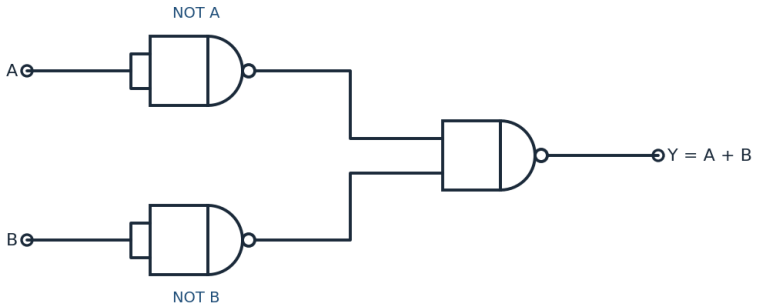
Counter / register	Facts
Asynchronous (ripple) counter	Each FF clocks the next; simple but delays add up; good for frequency division
Synchronous counter	All FFs share one clock; faster and glitch-free; more logic
Mod-N counter	Needs $n = \lceil \log_2 N \rceil$ FFs; $f_{out} = f_{clk}/N$
Ring counter	N FFs → N states, one-hot
Johnson counter	N FFs → 2N states, only one bit changes per step
Shift registers: SISO, SIPO, PISO, PIPO	SIPO converts serial to parallel (74HC595), PISO parallel to serial; delay lines, LFSR/CRC/PRBS

Device	Difference and application
Buffer	Non-inverting stage that restores logic levels and isolates source from load; used on clocks and long lines
Driver	Buffer with high current drive (LEDs, relays, buses); often tri-state or open-drain
Open-drain / open-collector	Pulls low only; needs a pull-up; allows wired-AND and level shifting (I^2C)
Analog comparator	Compares two analog voltages → digital output (threshold detect, zero crossing)
Digital comparator	Compares two binary numbers → A>B, A=B, A<B

7.2 Solved problems

Problem 7.1 – Build an OR gate from NAND gates

Question: Show how to implement Y = A + B using only NAND gates and prove it.



Solution

Step 1. A NAND with its inputs tied together is a NOT gate: NAND(A, A) = A′.

Step 2. Create A′ and B′ with two such gates.

Step 3. Feed them to a third NAND: Y = (A′·B′)′.

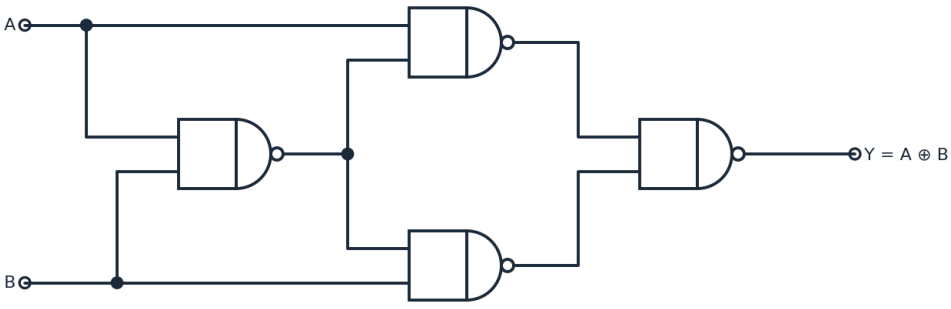
Step 4. By De Morgan's theorem (A′·B′)′ = A + B ✓. **Total: 3 NAND gates**.

Step 5. Truth check: A = 0, B = 0 → A′B′ = 1 → Y = 0 ✓; A = 1, B = 0 → A′B′ = 0 → Y = 1 ✓.

Answer: OR = NAND(NOT A, NOT B) – three NAND gates.

Problem 7.2 – XOR from four NAND gates

Question: Implement Y = A ⊕ B using four NAND gates and verify with a truth table.



Solution

Step 1. N1 = NAND(A, B). N2 = NAND(A, N1). N3 = NAND(B, N1). Y = NAND(N2, N3).

Step 2. Verify: A = 0, B = 0 → N1 = 1, N2 = 1, N3 = 1 → Y = 0 ✓.

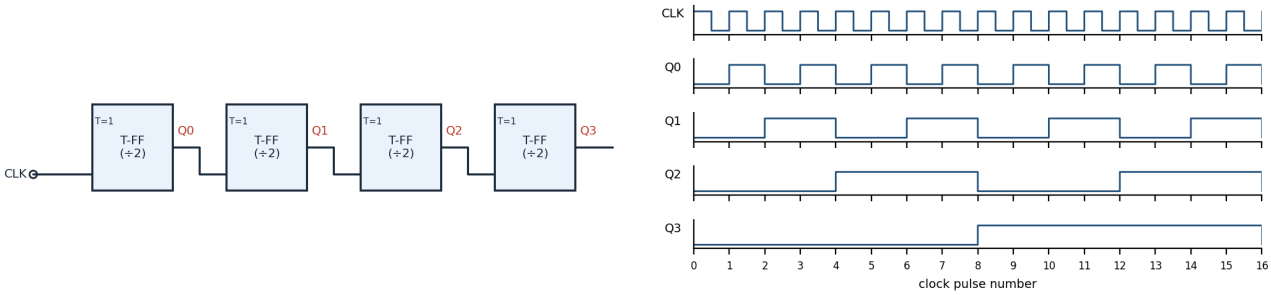
Step 3. A = 0, B = 1 → N1 = 1, N2 = NAND(0,1) = 1, N3 = NAND(1,1) = 0 → Y = NAND(1,0) = 1 ✓.

Step 4. A = 1, B = 0 → symmetric → Y = 1 ✓. A = 1, B = 1 → N1 = 0, N2 = N3 = 1 → Y = NAND(1,1) = 0 ✓.

Answer: Y = A ⊕ B; truth table 0, 1, 1, 0.

Problem 7.3 – Ripple counter and frequency division

Question: A 4-bit ripple counter built from toggle flip-flops is clocked at 32 MHz. Find the frequency at Q0, Q1, Q2 and Q3, the number of states, and the number of flip-flops needed for a mod-60 counter.



Solution

Step 1. Each toggle FF divides its clock by 2: Q0 = 32/2 = **16 MHz**, Q1 = **8 MHz**, Q2 = **4 MHz**, Q3 = **2 MHz** (= $f_{clk}/2^4 = f/16$).

Step 2. A 4-bit counter has 2^4 = **16 states** (0 to 15) and counts up on each clock pulse as shown in the timing diagram.

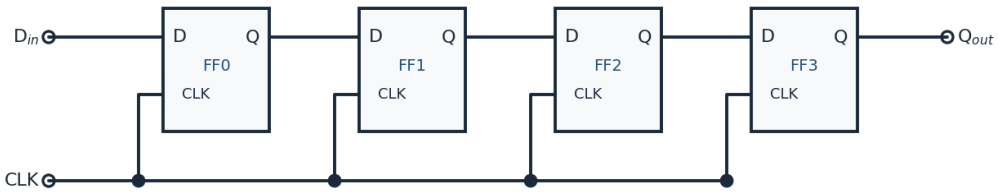
Step 3. Mod-60: need $2^n \geq 60 \rightarrow n =$ **6 flip-flops** (64 states), with logic that resets the counter at count 60.

Step 4. Because each FF waits for the previous one, ripple counters have accumulated delay (= $n \times t_{pd}$) and brief glitches; synchronous counters avoid this.

Answer: 16, 8, 4, 2 MHz; 16 states; 6 FFs for mod-60.

Problem 7.4 – Serial-in shift register

Question: A 4-bit SISO shift register (initially 0000) receives the serial data 1, 0, 1, 1 (first bit first), one bit per clock. Show the register contents after each clock and state how long it takes at 1 MHz.



Solution

Step 1. Each clock edge moves every bit one place to the right ($Q_0 \rightarrow Q_1 \rightarrow Q_2 \rightarrow Q_3$) and loads the new serial bit into FF0.

Step 2. Contents ($Q_0\ Q_1\ Q_2\ Q_3$) after each clock: **clock 1: 1000 → clock 2: 0100 → clock 3: 1010 → clock 4: 1101**.

Step 3. After 4 clocks the first bit sent has reached the last flip-flop, so reading $Q_3\ Q_2\ Q_1\ Q_0$ gives **1011** – exactly the word that was sent.

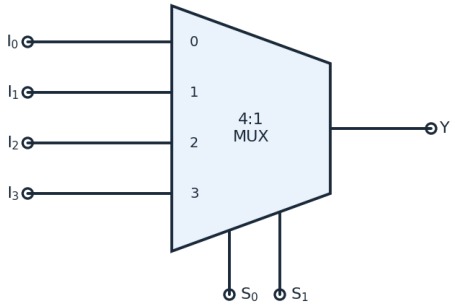
Step 4. Time = 4 clocks × 1 µs = **4 µs** (n bits need n clock pulses; the same again to shift them out serially).

Step 5. Reading all FFs at once (SIPO) converts serial data to parallel – e.g. the 74HC595 expands GPIO pins.

Answer: Contents: 1000, 0100, 1010, 1101; 4 µs total.

Problem 7.5 – Implementing a function with a multiplexer

Question: Implement F(A,B,C) = Σm(1, 2, 6, 7) using one 4:1 multiplexer, with A and B as select lines (A = S_1, B = S_0).



Solution

Step 1. Minterm index = 4A + 2B + C. Group the minterms in pairs by (A,B):

Step 2. AB = 00 → m0, m1: only m1 is in F → F = C → $\mathbf{I_0 = C}$.

Step 3. AB = 01 → m2, m3: only m2 → F = C′ → $\mathbf{I_1 = C'}$.

Step 4. AB = 10 → m4, m5: none → $\mathbf{I_2 = 0}$. AB = 11 → m6, m7: both → $\mathbf{I_3 = 1}$.

Step 5. Wire I_0 = C, I_1 = C′ (one inverter), I_2 = GND, I_3 = V_{DD}. A 2^n:1 mux implements any (n+1)-variable function this way.

Answer: I_0 = C, I_1 = C′, I_2 = 0, I_3 = 1.

Quick concept questions

Q. How many flip-flops count 100 states? Johnson counter states for 6 FFs?

A. $2^7 = 128 \geq 100$ → **7 FFs**; Johnson: 2 × 6 = **12 states**.

Q. TTL levels: V_{OH} = 2.4 V, V_{IH} = 2.0 V, V_{OL} = 0.4 V, V_{IL} = 0.8 V. Noise margins?

A. NM_H = 2.4 − 2.0 = 0.4 V; NM_L = 0.8 − 0.4 = 0.4 V.

Q. Why must an open-collector output have a pull-up?

A. It can only pull the line low; the resistor provides the high level.

Q. Simplify F = AB + AB′ + A′B.

A. AB + AB′ = A → F = A + A′B = **A + B**.

8. Digital IC Characteristics – Timing Parameters

In one line: a digital circuit works only if every signal arrives at a flip-flop **early enough (setup)** and stays **long enough (hold)** around the clock edge. These timing rules decide the maximum clock frequency.

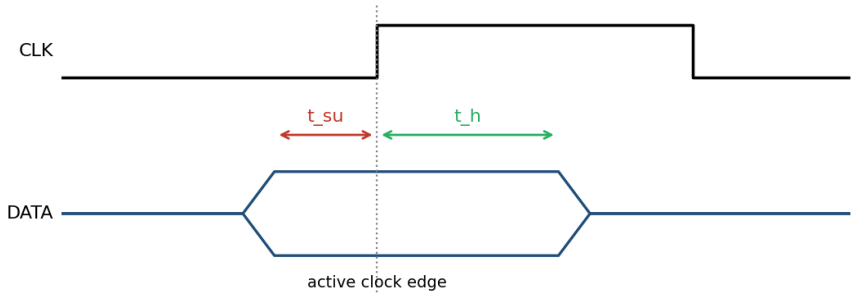


Fig 8.1 – Setup time (before the clock edge) and hold time (after it).

8.1 Timing parameters – definition and meaning

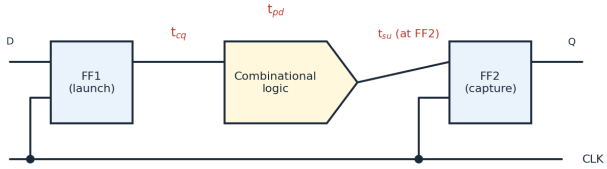
Parameter	Definition	Why it matters
Setup time t_{su}	Data must be stable this long **before** the active clock edge	Limits max frequency
Hold time t_h	Data must stay stable this long **after** the clock edge	Not fixed by slowing the clock
Clock-to-Q t_{cq}	Delay from clock edge to valid output	Part of the path delay
Propagation delay t_{pd}	Input-to-output delay measured at 50 % points (t_{pLH}, t_{pHL})	Logic depth × delay
Rise / fall time	10 % → 90 % (or 20 → 80 %) of the swing	Edge speed, signal integrity
Bandwidth	BW ≈ $0.35/t_r$	Relates edge speed to scope/probe needs
Skew / jitter	Skew: clock arrival difference between FFs; jitter: edge variation cycle to cycle	Eats timing margin
Metastability	FF output stuck between levels when setup/hold is violated	Use 2-FF synchronisers for asynchronous inputs

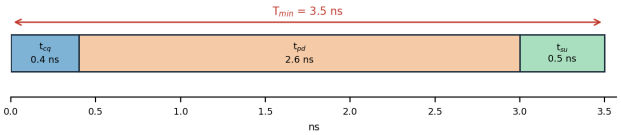
Key formulae	
Setup check	$T_{clk} \geq t_{cq} + t_{pd,max} + t_{su} + t_{skew}$; $f_{max} = 1/T_{min}$
Setup slack	$Slack = T_{clk} - (t_{cq} + t_{pd,max} + t_{su} + t_{skew}) \geq 0$
Hold check	$t_{cq,min} + t_{pd,min} \geq t_h + t_{skew}$ (independent of frequency)
Edge / bandwidth	$t_r = 0.35/BW$; $t_{r,meas} = \sqrt{(t_{r,sig}^2 + t_{r,scope}^2)}$
Duty cycle	$Duty = t_{high}/T \times 100\ \%$; $f = 1/T$

8.2 Solved problems

Problem 8.1 – Maximum clock frequency

Question: Two flip-flops are separated by combinational logic. t_{cq} = 0.4 ns, $t_{pd,max}$ = 2.6 ns, t_{su} = 0.5 ns, no skew. Find f_{max}, and the setup slack when the clock is 200 MHz.





Solution

Step 1. The data launched by FF1 at the clock edge must reach FF2 and settle *before* the next edge: $T_{min} = t_{cq} + t_{pd} + t_{su} = 0.4 + 2.6 + 0.5 =$ **3.5 ns**.

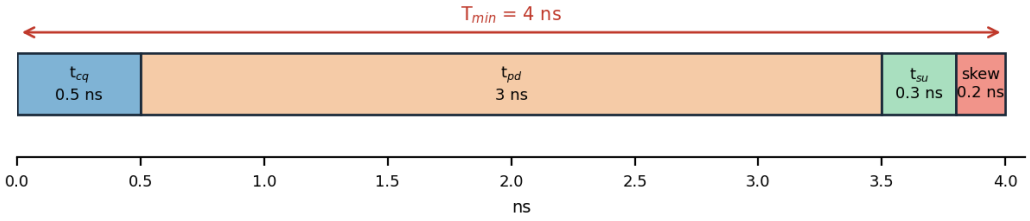
Step 2. $f_{max} = 1/3.5$ ns = **285.7 MHz**.

Step 3. At 200 MHz: T = 5 ns → slack = 5 − 3.5 = **+1.5 ns** (positive = timing met).

Answer: f_{max} = 285.7 MHz; slack at 200 MHz = +1.5 ns.

Problem 8.2 – Including clock skew

Question: t_{cq} = 0.5 ns, t_{pd} = 3 ns, t_{su} = 0.3 ns and the capture clock arrives 0.2 ns early (skew). Find f_{max} and the needed t_{pd} reduction to run at 300 MHz.



Solution

Step 1. $T_{min} = 0.5 + 3.0 + 0.3 + 0.2 =$ **4.0 ns** → $f_{max} =$ **250 MHz**.

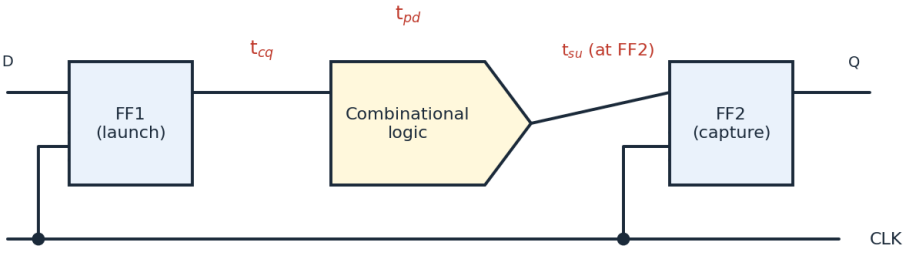
Step 2. 300 MHz → T = 3.33 ns. Fixed overhead = 0.5 + 0.3 + 0.2 = 1.0 ns, so $t_{pd} \le 2.33$ ns.

Step 3. Required reduction = 3.0 − 2.33 = **0.67 ns** (e.g. pipeline the logic or reduce its depth).

Answer: f_{max} = 250 MHz; cut t_{pd} by 0.67 ns for 300 MHz.

Problem 8.3 – Hold-time violation

Question: $t_{cq,min}$ = 0.3 ns, $t_{pd,min}$ = 0.2 ns, t_h = 0.4 ns, skew = 0.2 ns. Does the path meet hold? Does slowing the clock fix it?



Solution

Step 1. Data arrives at the earliest after $t_{cq,min} + t_{pd,min} = 0.3 + 0.2 = \mathbf{0.5\ ns}$.

Step 2. It must not change before t_h + skew = 0.4 + 0.2 = **0.6 ns**.

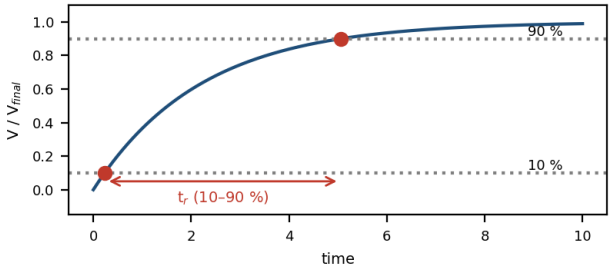
Step 3. 0.5 < 0.6 → **hold violation by 0.1 ns** (new data races through and corrupts the capture).

Step 4. Slowing the clock does **not** help: hold is checked against the *same* clock edge. Fix by adding delay (buffers) in the data path of ≥ 0.1 ns or reducing the skew.

Answer: Violation of 0.1 ns – fix by inserting delay in the data path.

Problem 8.4 – Rise time, bandwidth and scope accuracy

Question: A logic edge has t_r = 1 ns. (a) What signal bandwidth does it need? (b) A 200 MHz scope (t_r = 1.75 ns) is used – what rise time will it display?



Solution

Step 1. (a) BW = $0.35/t_r$ = 0.35/1 ns = **350 MHz**.

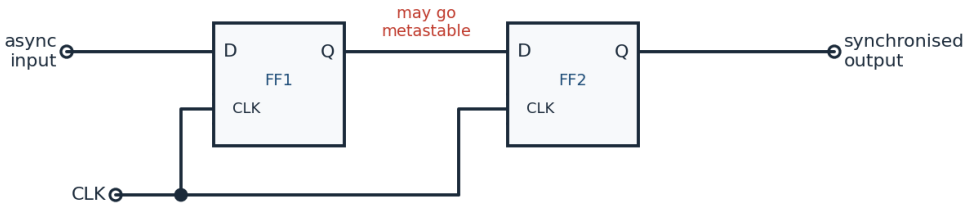
Step 2. (b) Displayed $t_r = \sqrt{(1^2 + 1.75^2)} = \sqrt{(1 + 3.06)} =$ **2.02 ns** – twice the true value!

Step 3. Lesson: choose a scope bandwidth at least 3–5× the signal bandwidth (here ≥ 1 GHz) so the instrument does not distort the measurement.

Answer: Signal needs 350 MHz; the 200 MHz scope shows 2.02 ns.

Problem 8.5 – Metastability and the two-flip-flop synchroniser

Question: An asynchronous push-button signal enters a clocked design. Why is a single flip-flop unsafe, and how does the circuit below help?



Solution

Step 1. An asynchronous input can change at any time, including inside the setup/hold window of FF1. Then FF1's output may hover at an **intermediate level** for an unpredictable time (metastability).

Step 2. FF2 samples FF1 one clock later. By then the metastable output has almost always resolved to 0 or 1, so the logic after FF2 sees a clean level.

Step 3. MTBF = $e^{t_r/\tau}/(T_0 f_{clk} f_{data})$ – it grows **exponentially** with the extra resolution time t_r (one clock period). For a faster clock or slower FFs add a third stage.

Step 4. Never use FF1's output directly, and never cross clock domains with multi-bit buses this way (use FIFO or Gray code).

Answer: A two-stage synchroniser trades one clock of latency for a much longer MTBF.

Quick concept questions

Q. Duty cycle and frequency of a clock high for 30 ns in a 100 ns period?

A. Duty = **30 %**; f = 1/100 ns = **10 MHz**.

Q. Does a hold violation depend on clock frequency?

A. No – only on data-path delay versus hold requirement plus skew.

9. ADC and DAC – Static and Dynamic Parameters

In one line: an **ADC** converts a continuous voltage into an N-bit number; a **DAC** does the reverse. The step size is one **LSB** = $V_{ref}/2^N$, and every converter has errors that are described by *static* (DC) and *dynamic* (AC) specifications.

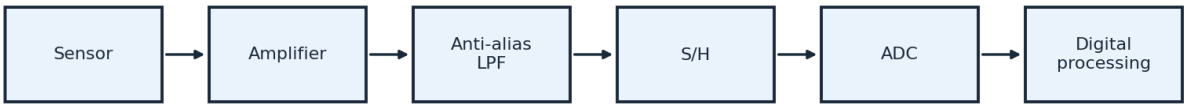


Fig 9.1 – Typical data-acquisition chain: the filter prevents aliasing, the S/H freezes the input while the ADC converts.

9.1 Types of ADCs

ADC type	Speed / resolution	How it works / use
Flash (parallel)	Fastest (GSPS); 6–8 bit	$2^N - 1$ comparators; size and power grow rapidly; oscilloscopes, video
SAR (successive approximation)	kSPS–10s MSPS; 8–18 bit	Binary search with a DAC, N clocks per sample; most microcontroller ADCs
Pipeline	10–500 MSPS; 8–16 bit	Several stages each resolve a few bits; latency of a few clocks
Sigma-delta (Σ-Δ)	Slow; 16–24+ bit	Oversampling + noise shaping + digital filter; audio, sensors, weigh scales
Dual-slope (integrating)	Very slow, accurate	Integrates input then a reference; rejects noise; multimeters

9.2 Types of DACs and converter specifications

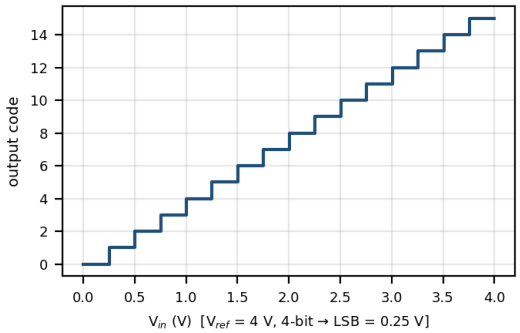
Item	Description	Note
Binary-weighted DAC	Resistors R, 2R, 4R … per bit	Wide resistor spread – hard to match
R-2R ladder DAC	Only two resistor values	Good matching; very common
Sigma-delta / PWM DAC	Bit-stream + low-pass filter	Audio, high resolution
Static specs	Offset, gain error, **INL**, **DNL**, monotonicity, missing codes	Measured with slow ramp / code-density histogram
Dynamic specs	SNR, SINAD, THD, SFDR, **ENOB**, settling time, glitch energy, aperture jitter, bandwidth	Measured with a sine wave + FFT

Key formulae	
LSB / codes	LSB = $V_{FS}/2^N$ ($\approx V_{ref}/2^N$); number of codes = 2^N
ADC code	$D = \text{floor}(V_{in}/\text{LSB})$
DAC output	$V_{out} = V_{ref} \cdot D/2^N$
Quantisation	error ±½ LSB; noise rms = LSB/√12
Ideal SNR	SNR = 6.02N + 1.76 dB
ENOB	ENOB = (SINAD − 1.76)/6.02
Nyquist	$f_s \geq 2f_{max}$; higher frequencies alias
DNL / INL	DNL = (step width/LSB) − 1; INL = cumulative deviation from ideal line; DNL = −1 LSB ⇒ missing code
Jitter limit	SNR = $-20 \log(2\pi f_{in} t_j)$
Oversampling	4× oversampling ≈ +6 dB ≈ +1 bit

9.3 Solved problems

Problem 9.1 – Resolution and quantisation of a 4-bit ADC

Question: A 4-bit ADC has V_{ref} = 4 V. (a) Find the LSB. (b) What code does 2.9 V give? (c) What is the maximum quantisation error?



Solution

Step 1. Number of codes = 2^4 = 16 (0 … 15). LSB = 4 V/16 = **0.25 V**.

Step 2. Code = floor(2.9/0.25) = floor(11.6) = **11** = binary **1011** (covers the range 2.75–3.00 V).

Step 3. The converter reports the centre value 2.875 V; the error is ≤ ±½ LSB = **±0.125 V**.

Step 4. Each extra bit halves the LSB: at 12 bits the same range gives 0.98 mV steps.

Answer: LSB = 0.25 V; code 1011 (11); max error ±0.125 V.

Problem 9.2 – 12-bit ADC – LSB, code, SNR and ENOB

Question: A 12-bit ADC has V_{ref} = 3.3 V. Find the LSB, the code for 1.2 V, the ideal SNR, and the ENOB if the measured SINAD is 68 dB.

Solution

Step 1. LSB = 3.3/4096 = 0.806 mV → **0.806 mV**.

Step 2. Code = 1.2/0.000806 = 1489.5 → **1489** (0x5D1).

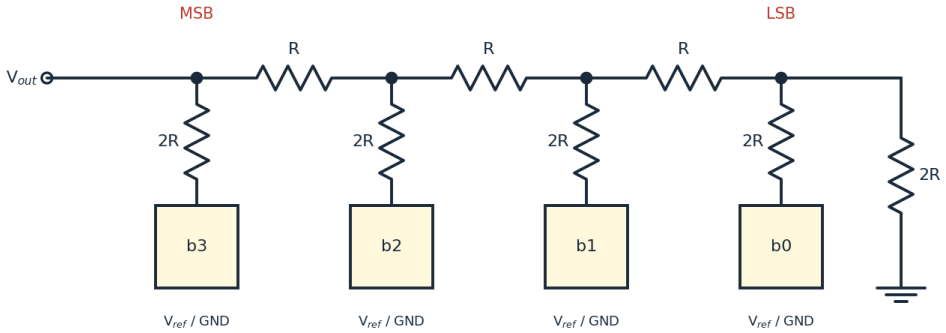
Step 3. Ideal SNR = 6.02 × 12 + 1.76 = **74 dB**.

Step 4. ENOB = (68 − 1.76)/6.02 = **11.0 bits** – the real converter behaves like an 11-bit ideal one (noise and distortion cost 1 bit).

Answer: LSB = 0.806 mV; code 1489; SNR 74 dB; ENOB = 11.0 bits.

Problem 9.3 – R-2R ladder DAC

Question: In the 4-bit R-2R DAC each bit switch connects to V_{ref} = 5 V (bit = 1) or to ground (bit = 0). Find V_{out} for the code 1010 (b3 b2 b1 b0) and the LSB step.



Solution

Step 1. At each node of the ladder the resistance looking to the right equals 2R, so every bit contributes half the previous bit's weight: b3 → $V_{ref}/2$, b2 → $V_{ref}/4$, b1 → $V_{ref}/8$, b0 → $V_{ref}/16$.

Step 2. Weights for 5 V: 2.5 V, 1.25 V, 0.625 V, 0.3125 V.

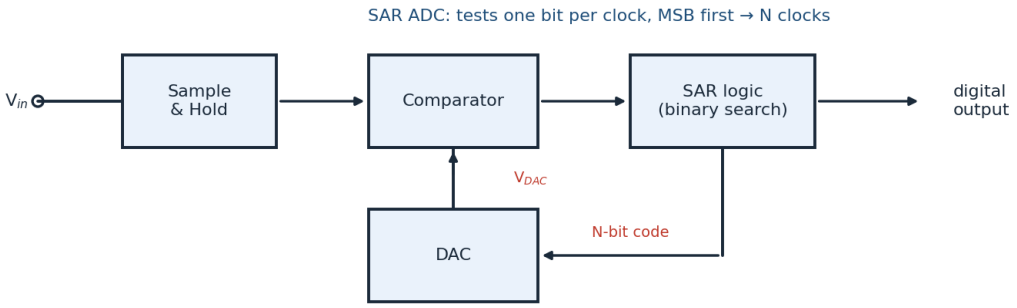
Step 3. Code 1010: b3 = 1, b1 = 1 → V_{out} = 2.5 + 0.625 = **3.125 V** (= 5 × 10/16 ✓).

Step 4. LSB = $V_{ref}/16$ = **0.3125 V**; full scale (1111) = 4.6875 V = one LSB below V_{ref}.

Answer: V_{out} = 3.125 V; LSB = 0.3125 V.

Problem 9.4 – SAR conversion step by step

Question: A 4-bit SAR ADC with V_{ref} = 5 V converts V_{in} = 3.3 V. Show each comparison.



Solution

Step 1. The SAR logic tries each bit from the MSB, setting it to 1, asking the DAC for V_{DAC} and keeping the bit only if $V_{DAC} \le V_{in}$.

Step 2. Trial 1: 1000 → V_{DAC} = 5 × 8/16 = 2.5 V ≤ 3.3 → keep b3 = **1**.

Step 3. Trial 2: 1100 → 3.75 V > 3.3 → drop, b2 = **0**.

Step 4. Trial 3: 1010 → 3.125 V ≤ 3.3 → keep, b1 = **1**.

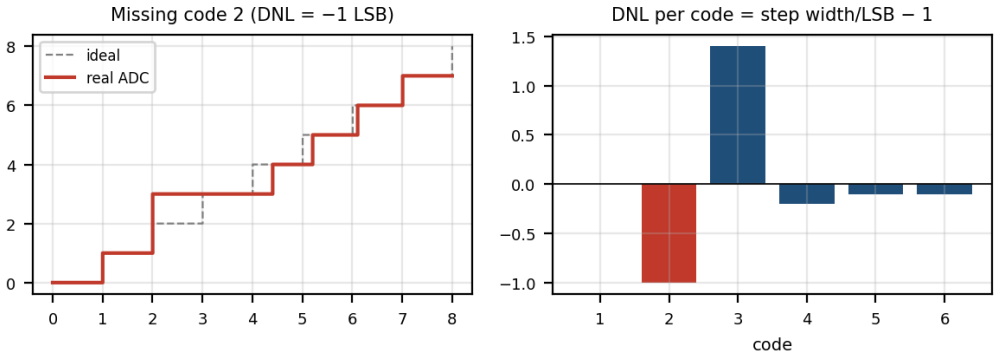
Step 5. Trial 4: 1011 → 3.4375 V > 3.3 → drop, b0 = **0**.

Step 6. Result = **1010** = 10 (check: 3.3/0.3125 = 10.56 → 10 ✓). A SAR converter always needs exactly N comparisons – N clock cycles.

Answer: Output code 1010 after 4 clock cycles.

Problem 9.5 – Reading DNL and INL – missing codes

Question: The figure shows a real 3-bit ADC with its DNL plot. Which code is missing and which is too wide? Find the DNL values and the INL at code 3.



Solution

Step 1. DNL of a code = (width of its step in LSB) − 1.

Step 2. Code 1: width 1.0 LSB → DNL = **0**. Code 2: width 0 → DNL = **−1 LSB** → this is a **missing code** (the output jumps from 1 to 3).

Step 3. Code 3: width 2.4 LSB → DNL = **+1.4 LSB** (too wide, picks up the missing step). Codes 4–6 are slightly narrow (−0.2, −0.1, −0.1).

Step 4. INL = running sum of DNL: after code 1: 0, after code 2: −1, after code 3: −1 + 1.4 = **+0.4 LSB** – the transfer line is only 0.4 LSB away from the ideal line even though a code is missing.

Step 5. Acceptance rule: |DNL| < 1 LSB and monotonic output ⇒ no missing codes.

Answer: Code 2 is missing (DNL = −1 LSB); code 3 is too wide (+1.4 LSB); INL(3) = +0.4 LSB.

Problem 9.6 – Sampling rate and clock jitter

Question: (a) What minimum sample rate is needed for 20 kHz audio? (b) A 10 MHz input is sampled with 1 ps rms clock jitter. What is the best possible SNR and the equivalent number of bits?

Solution

Step 1. (a) Nyquist: $f_s > 2 \times 20$ kHz = **40 kHz** (CD uses 44.1 kHz so the anti-alias filter can roll off between 20 and 22 kHz).

Step 2. (b) $SNR_{jitter} = -20 \log(2\pi f_{in} t_j) = -20 \log(2\pi \times 10^{7} \times 10^{-12}) = -20 \log(6.28 \times 10^{-5})$ = **84 dB**.

Step 3. ENOB = (84 − 1.76)/6.02 = **13.7 bits** – no ADC with more than ≈ 14 bits is useful at 10 MHz with this clock.

Answer: (a) > 40 kHz; (b) 84 dB ≈ 13.7 bits.

Quick concept questions

Q. Flash ADC: how many comparators for 8 bits? SAR: how many clock cycles for 12 bits?

A. Flash: $2^8 - 1$ = **255**. SAR: **12** comparisons (one per bit).

Q. To gain 2 extra bits by plain oversampling, by how much must you oversample?

A. Each bit needs 4× → 4^2 = **16×**.

Q. 8-bit DAC, 2.56 V reference, input 0x80. Output?

A. 128/256 × 2.56 = **1.28 V**.

10. Analog vs Digital – Modes, Differences and Interfaces

In one line: analog signals vary continuously and carry information in their exact value; digital signals use a few levels, so noise can be removed completely.

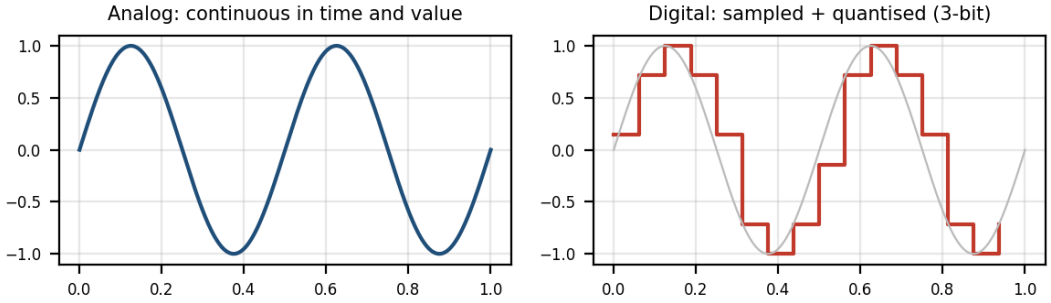


Fig 10.1 – A continuous analog signal and its sampled, quantised digital version.

	Analog	**Digital**
Signal	Continuous in time and amplitude	Discrete levels (0/1) sampled in time
Noise immunity	Poor – noise stays in the signal	Excellent – noise margin lets the receiver regenerate exact bits
Accuracy	Limited by components, drift, temperature	Set by word length; error correction possible
Storage / processing	Hard; degrades on copying	Easy, lossless, programmable (MCU, DSP, FPGA)
Power / area	Scales poorly in advanced processes	Scales well (Moore's law)
Examples	Op-amp, sensor, LDO, PLL, RF	Gates, microcontroller, memory, FPGA

Mixed-signal chips (ADC, DAC, PLL, PMIC, SerDes, sensors) contain both; validation therefore needs analog characterisation (V, T sweeps) *and* digital functional testing.

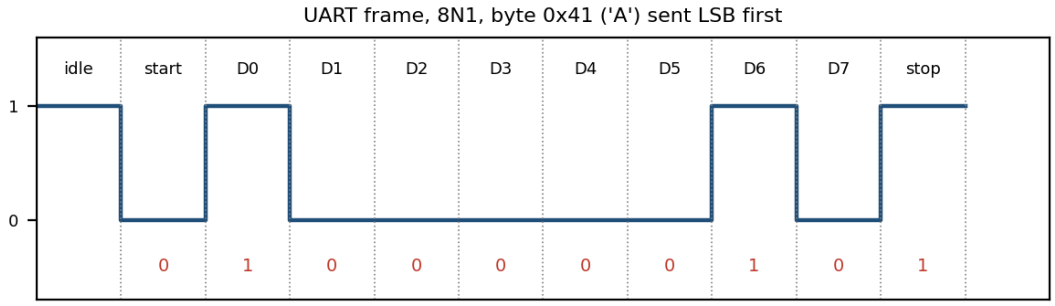
10.1 Common digital interfaces

Interface	**Wires**	**Key facts**
UART	TX, RX (asynchronous)	No clock – both sides agree a baud rate; 8N1 frame = start + 8 data (LSB first) + stop = 10 bits
SPI	SCLK, MOSI, MISO, CS	Synchronous, full duplex, fast (tens of MHz); master selects slave with CS; modes 0–3 (CPOL/CPHA)
I²C	SDA, SCL (open-drain)	Addressed multi-device bus; needs pull-ups; 100/400/1000 kHz; ACK after each byte
GPIO / PWM	1 pin	Digital I/O; PWM average voltage = duty × V_{DD}
CAN, USB, Ethernet	Differential	Robust / high-speed serial links (SerDes)

10.2 Solved problems

Problem 10.1 – Decoding a UART frame and its timing

Question: The waveform shows a UART frame (8N1, LSB first). (a) Decode the byte. (b) How long does one byte take at 115200 baud, and how many bytes per second? (c) How long to send 1 KB?



Solution

Step 1. (a) The line idles high. Start bit = 0. Data bits D0 … D7 = **1, 0, 0, 0, 0, 0, 1, 0**. Stop = 1.

Step 2. UART sends the LSB first, so reading D7 … D0 gives 0100 0001 = **0x41 = 'A'**.

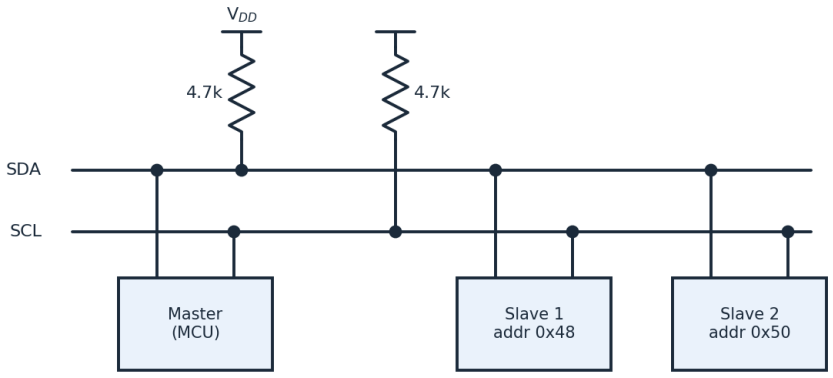
Step 3. (b) Bit time = 1/115200 = **8.68 µs**. A frame has 1 + 8 + 1 = 10 bits → 86.8 µs per byte → **11 520 bytes/s**.

Step 4. (c) 1024 bytes × 86.8 µs = **88.9 ms** (at 9600 baud the same transfer would take 1.07 s).

Answer: Byte 0x41 ('A'); 86.8 µs per byte; 11 520 B/s; 1 KB in 88.9 ms.

Problem 10.2 – Choosing I²C pull-up resistors

Question: An I²C bus (V_{DD} = 3.3 V) runs at 400 kHz with a total bus capacitance of 100 pF. The maximum allowed rise time is 300 ns and the devices can sink 3 mA at V_{OL} = 0.4 V. Find the allowed range of the pull-up resistor.



Solution

Step 1. Why pull-ups? Devices only pull the lines low (open-drain), so a resistor must pull them high; this also lets many devices share the lines without conflict.

Step 2. Rise time: $t_r \approx 0.8473 \times R_p \times C_b \le 300$ ns → $R_{p,max}$ = 300 ns/(0.8473 × 100 pF) = **3.54 kΩ**.

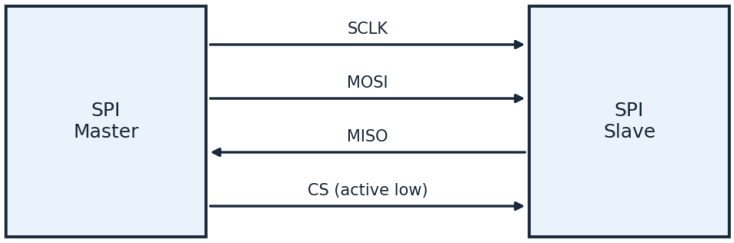
Step 3. Sink current: $R_{p,min} = (V_{DD} - V_{OL})/I_{OL}$ = (3.3 − 0.4)/3 mA = **967 Ω**.

Step 4. Choose between 1 kΩ and 3.5 kΩ, e.g. **2.2 kΩ** (a good compromise between speed and power).

Answer: R_p between 0.97 kΩ and 3.54 kΩ; 2.2 kΩ is a good choice.

Problem 10.3 – SPI transfer time

Question: An SPI master clocks at 8 MHz and reads a 16-bit sensor register. How long does one transfer take and what is the maximum raw data rate? Which lines carry the data in each direction?



Solution

Step 1. Clock period = 1/8 MHz = 125 ns. 16 bits need 16 clocks → **2 µs** (plus a little CS set-up time).

Step 2. Raw rate = 8 Mbit/s = **1 MB/s**, in both directions at the same time (full duplex).

Step 3. Master → slave data on **MOSI**; slave → master data on **MISO**; **CS** (active low) selects the device and frames the transfer; **SCLK** is generated by the master.

Step 4. CPOL/CPHA (modes 0–3) define the idle level of the clock and which edge samples data – both sides must use the same mode.

Answer: 2 µs per 16-bit transfer; 8 Mbit/s.

Problem 10.4 – Why digital survives noise (noise margin)

Question: A 3.3 V logic line (V_{OH} = 3.3 V, V_{IH} = 2.0 V, V_{IL} = 0.8 V) and a 3.3 V analog level both pick up 0.5 V of noise. Compare the effect on the receiver.

Solution

Step 1. Digital: the HIGH level drops to 3.3 − 0.5 = 2.8 V, still above V_{IH} = 2.0 V → the receiver reads a clean logic 1 and re-drives a perfect 3.3 V. Noise margin NM_H = 3.3 − 2.0 = 1.3 V, so up to 1.3 V of noise is tolerated and **removed**.

Step 2. Analog: the receiver cannot tell signal from noise: the measured value is 2.8 V instead of 3.3 V – a permanent error of 15 %, and each further stage adds more noise.

Step 3. That is why long-distance links, memories and computers are digital, and why analog signals are digitised as close to the sensor as possible.

Answer: Digital: error-free (margin 1.3 V). Analog: 15 % permanent error.

Quick concept questions

Q. Why are I²C lines open-drain?

A. Many devices share a line; open-drain + pull-up avoids driver conflicts (wired-AND) and allows clock stretching.

Q. PWM at 3.3 V and 25 % duty – average voltage?

A. **0.825 V**.

Q. Name an analog block that stays analog in an all-digital SoC.

A. PLL/clock generator, I/O pads, ADC/DAC, voltage regulators, temperature sensors.

11. C Programming and Logical Thinking

In one line: validation engineers write C to access registers, manipulate bits and automate tests. Interviews check bit operations, pointers, memory layout and clear logic.

11.1 Key topics – types and meaning

Topic	Types / forms	What to remember
Data types	char 8-bit, short 16, int 32, long, float, double; `uint8_t, uint32_t` (stdint.h)	Use fixed-width types for hardware. Signed overflow is undefined; unsigned wraps around
Storage classes	`auto, static, extern, register`; qualifiers `const, volatile`	static local keeps its value between calls; **volatile** stops the compiler caching a register or ISR-shared variable
Operators	Arithmetic, relational, logical (&& || !), **bitwise** (& | ^ ~ << >>)	&& is logical, & is bitwise; << n multiplies by 2^n
Pointers	Data pointer, function pointer, NULL, void*, pointer to pointer	p + 1 moves sizeof(*p) bytes; arr[i] ≡ *(arr+i)
Aggregates	array, struct, union, enum, bit-field	struct adds padding; union shares memory; endianness = byte order
Memory areas	text, data, bss, heap, stack	Locals on the stack, malloc on the heap, globals in data/bss
Preprocessor	#define, #include, #ifdef, macros	Parenthesise macro arguments; use include guards

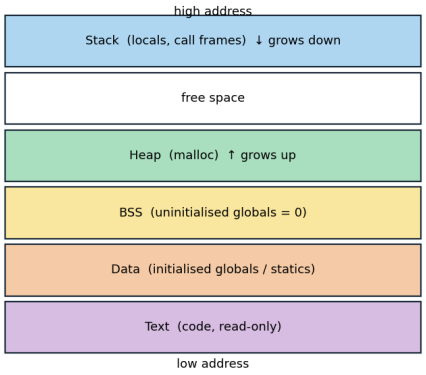


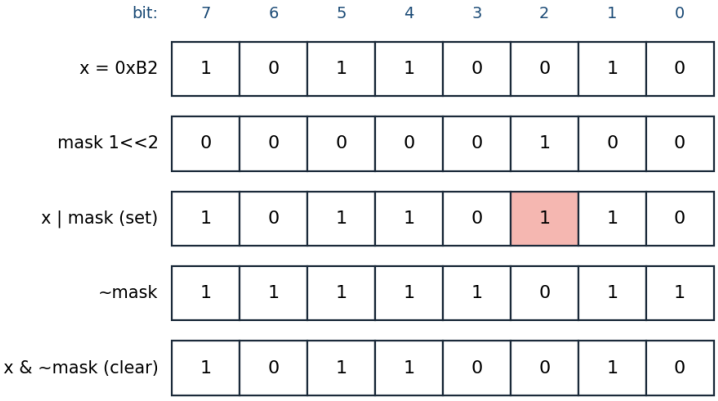
Fig 11.1 – Typical C memory layout.

11.2 Solved problems (with code)

Problem 11.1 – Set, clear, toggle and test a bit

Question: A control register holds x = 0xB2. Write C statements to (a) set bit 2, (b) clear bit 7, (c) toggle bit 0, (d) test bit 4 and give the results.

```
x |=  (1u << 2);      // set bit 2
x &= ~(1u << 7);      // clear bit 7
x ^=  (1u << 0);      // toggle bit 0
int b = (x >> 4) & 1; // read bit 4
```



Solution

Step 1. 0xB2 = 1011 0010 (bit 7 … bit 0). A mask `1u << n` has a single 1 at bit n.

Step 2. (a) Set: `x |= (1u << 2);` → 1011 0110 = **0xB6** (OR with 1 forces a 1).

Step 3. (b) Clear: `x &= ~(1u << 7);` → 0011 0010 = **0x32** (AND with the inverted mask forces a 0).

Step 4. (c) Toggle: `x ^= (1u << 0);` → 1011 0011 = **0xB3** (XOR with 1 flips the bit).

Step 5. (d) Test: `(x >> 4) & 1u` → bit 4 of 0xB2 is **1**.

Answer: 0xB6, 0x32, 0xB3 and bit 4 = 1.

Problem 11.2 – Write a bit-field in a register

Question: A register R = 0x5C. Set bits [7:4] to 0xA without disturbing the other bits. Give the C statement and the final value.

```
R = (R & ~(0xFu << 4)) | (0xAu << 4);
```

Solution

Step 1. Clear the field first: mask = 0xF << 4 = 0xF0; `R & ~0xF0` = 0x5C & 0x0F = 0x0C.

Step 2. Shift the new value into position: 0xA << 4 = 0xA0.

Step 3. OR them: 0x0C | 0xA0 = **0xAC**.

Step 4. General form: `R = (R & ~(mask << pos)) | ((val & mask) << pos);` – use this whenever you modify part of a hardware register (read-modify-write).

Answer: R = 0xAC.

Problem 11.3 – Count the set bits (Kernighan's method)

Question: Explain what `x & (x - 1)` does and trace the function for x = 0x2C.

```
int count_ones(uint32_t x) {
    int c = 0;
    while (x) { x &= (x - 1); c++; }
    return c;
}
```

Solution

Step 1. x − 1 flips the lowest set bit to 0 and turns all bits below it into 1s, so x & (x−1) **clears the lowest set bit**.

Step 2. x = 0x2C = 0010 1100. Iteration 1: 0x2C & 0x2B = 0x28 (0010 1000), count = 1.

Step 3. Iteration 2: 0x28 & 0x27 = 0x20 (0010 0000), count = 2.

Step 4. Iteration 3: 0x20 & 0x1F = 0x00, count = 3 → loop ends.

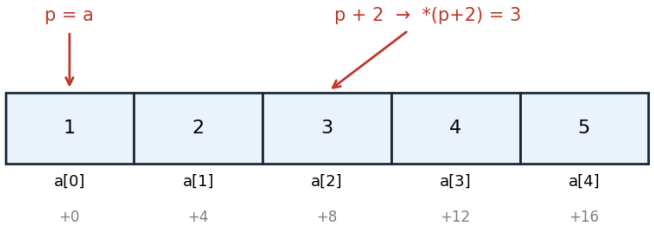
Step 5. The loop runs once per set bit (fast). The same trick tests a power of two: `x && !(x & (x-1))`.

Answer: Three set bits → returns 3.

Problem 11.4 – Pointers and arrays

Question: What does the program print? How many bytes is p + 2 beyond p?

```
int a[5] = {1, 2, 3, 4, 5};
int *p = a;
printf("%d ", *(p + 2));
p++;
printf("%d ", *p);
printf("%d", (int)(&a[4] - &a[1]));
```



Solution

Step 1. a is {1,2,3,4,5}; p = a points to a[0].

Step 2. *(p + 2) is the element two places on = a[2] = **3**. Pointer arithmetic scales by the element size: p + 2 is 2 × sizeof(int) = **8 bytes** further (for a 4-byte int).

Step 3. After p++ p points to a[1], so *p = **2**.

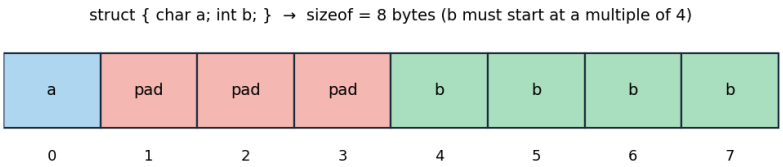
Step 4. &a[4] - &a[1] subtracts pointers and gives the number of *elements* between them = **3** (not bytes).

Answer: Output: 3 2 3.

Problem 11.5 – Structure padding

Question: On a 32-bit system, find sizeof for the two structures below and explain how to reduce the size.

```
struct S1 { char a; int b; };
struct S2 { char a; int b; char c; };
```



Solution

Step 1. The compiler aligns each member to a multiple of its own size. The int must start at a multiple of 4.

Step 2. S1: a at offset 0, 3 bytes of padding, b at 4–7 → **8 bytes**.

Step 3. S2 {char a; int b; char c;}: a (0), padding 1–3, b (4–7), c (8), padding 9–11 so the total is a multiple of 4 → **12 bytes**.

Step 4. Reordering to {int b; char a; char c;} gives 4 + 1 + 1 + 2 padding = **8 bytes**. Put large members first. Never assume sizeof(struct) = sum of members, especially when mapping a struct onto hardware registers.

Answer: S1 = 8 bytes, S2 = 12 bytes; reorder members to save space.

Problem 11.6 – Classic output-prediction traps

Question: Give the output of each statement: (a) unsigned char c = 255; c++; (b) #define SQR(x) x*x, value of SQR(2+3); (c) char c = -1; unsigned char u = c; (d) uint8_t v = ~0x0F.

Solution

Step 1. (a) 255 + 1 = 256 does not fit in 8 bits, so an unsigned value wraps around to **0**.

Step 2. (b) Macros do text substitution: 2+3*2+3 = 2 + 6 + 3 = **11**, not 25. Fix: `#define SQR(x) ((x)*(x))`.

Step 3. (c) −1 is 0xFF in two's complement; as unsigned char that is **255**.

Step 4. (d) ~0x0F = 0xFFFFFFF0 in an int, truncated to 8 bits = **0xF0 (240)**.

Answer: (a) 0; (b) 11; (c) 255; (d) 0xF0.

Problem 11.7 – Find the missing number with XOR

Question: An array holds the numbers 1 to 5 except one: {1, 2, 4, 5}. Find the missing number using XOR in O(n) time and O(1) memory.

```
int missing(int a[], int n) {      // a has n-1 values from 1..n
    int x = 0;
    for (int i = 1; i <= n; i++) x ^= i;
    for (int i = 0; i < n - 1; i++) x ^= a[i];
    return x;
}
```

Solution

Step 1. XOR has the properties a ^ a = 0 and a ^ 0 = a, and is order-independent.

Step 2. XOR all numbers 1...5: 1^2^3^4^5 = **1**.

Step 3. XOR the array elements into it: 1^1^2^4^5 = 0^2^4^5 = **3**.

Step 4. Every number that is present cancels itself, leaving the missing one: **3**. (The sum method works too, but can overflow.)

Answer: Missing number = 3.

Problem 11.8 – volatile and memory-mapped registers

Question: Why does the first loop below hang or be optimised away, and how do you write the second correctly? Where do the variables g, s, l and p live?

```
#define STATUS (*(volatile uint32_t *)0x40021000u)
while ((STATUS & 0x1) == 0) { }      // wait for READY bit

int g = 5;                           // global
void f(void) { static int s; int l; int *p = malloc(4); }
```

Solution

Step 1. A status register changes by itself (hardware). Without `volatile` the compiler may read it once, keep it in a CPU register and never re-read it, so the loop never ends.

Step 2. `volatile` tells the compiler: every access must really go to memory.

Step 3. Memory areas (Fig 11.1): **g** (initialised global) → data segment; **s** (static local) → data/bss and keeps its value; **l** (local) → stack; ***p** (malloc) → heap, while the pointer **p** itself is on the stack.

Step 4. Also: always free() what you malloc(), and never return the address of a local variable.

Answer: Use a volatile pointer to the register; g in data, l and p on the stack, *p on the heap.

Quick concept questions

Q. What is endianness? How is 0x12345678 stored on a little-endian machine?

A. Byte order in memory. Little-endian stores the least significant byte first: **78 56 34 12**.

Q. sizeof(int *) on a 32-bit vs 64-bit system?

A. 4 bytes vs 8 bytes.

Q. Difference between `const` and `volatile`?

A. const: the program may not write it. volatile: the value may change outside the program's control; always re-read it. They can be used together for read-only hardware registers.

12. Power Sources and Test & Measurement Equipment

In one line: every measurement depends on a clean power source and the right instrument used correctly. Know both the types and the typical mistakes.

12.1 Types of power sources

Source	Key facts	Typical use
Primary batteries (alkaline 1.5 V, Li coin 3 V)	Not rechargeable; very low self-discharge	Remote controls, RTC backup
Li-ion / Li-polymer	3.7 V nominal, 4.2 V full, ≈ 3.0 V empty; needs protection and CC/CV charging	Phones, laptops, portable devices
NiMH / NiCd	1.2 V per cell; rechargeable	Cameras, tools
Lead-acid	2 V/cell (12 V battery); heavy, cheap	UPS, vehicles
AC mains adapter / SMPS	Rectify + switch-mode conversion; 230 V, 50 Hz in India; efficient	Chargers, supplies
Linear bench supply	Very low noise, heavy	Analog and low-noise work
Programmable bench PSU	Constant-voltage (CV) and constant-current (CC) modes, OVP/OCP, remote sense	Lab bring-up, I_{DD} tests
USB / USB-PD, PoE, solar	5 V (PD up to 48 V); PoE via Ethernet; solar with MPPT	Portable and remote systems

12.2 Types of test and measurement instruments

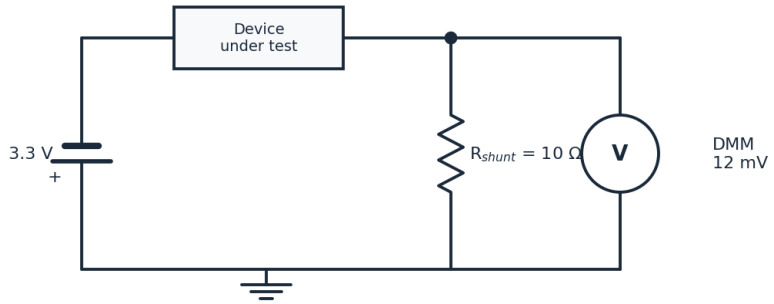
Instrument	Measures	Key points
Digital multimeter (DMM)	V, I, R, continuity, diode, capacitance	Voltage: in parallel; current: in **series**; resistance: power off; 4-wire (Kelvin) for mΩ
Oscilloscope	Voltage vs time; rise time, jitter, ripple	Bandwidth ≥ 3–5× signal; compensate 10× probe; short ground spring; correct trigger
Function / arbitrary generator	Sources sine, square, pulse	50 Ω output: set for high-Z load or terminate
Logic analyzer	Many digital channels, timing + protocol decode	I²C, SPI, UART debugging
Spectrum analyzer	Amplitude vs frequency	Harmonics, EMI, phase noise
SMU / source-measure unit	Sources V or I and measures the other	I–V curves, leakage, µA-level I_{DD}
Electronic load	Programmable sink (CC, CV, CR, CP)	Supply and battery testing
LCR meter / VNA	L, C, R; S-parameters	Component and RF characterisation
JTAG / protocol analyzer / debugger	On-chip and bus-level debug	Register peek/poke, boundary scan

Key formulae	
Battery	Runtime = capacity (Ah)/I (derate 10–30 %); Energy (Wh) = $V_{nom} \times$ Ah
Efficiency	$\eta = P_{out}/P_{in}$, $P_{in} = P_{out}/\eta$
Shunt measurement	$I = V_{shunt}/R_{shunt}$; burden voltage = $I \cdot R_{shunt}$
10× probe	R = 9 MΩ + scope 1 MΩ = 10 MΩ; attenuation 10×; input C ≈ 10–15 pF
Scope bandwidth	BW ≥ 5 × f_{signal}; $t_r = 0.35/BW$; sample rate ≥ 2.5–5 × BW
Ripple	ripple % = $V_{pp}/V_{DC} \times 100$

12.3 Solved problems

Problem 12.1 – Measuring current with a shunt resistor

Question: A DMM reads 12 mV across a 10 Ω shunt in series with a device. Find the current. If the device draws up to 100 mA and the burden must stay below 100 mV, what shunt value should be used and what voltage does 1.2 mA then give?



Solution

Step 1. I = V/R = 12 mV/10 Ω = **1.2 mA**. The shunt also drops 12 mV from the device's supply (burden voltage).

Step 2. For 100 mA max with ≤ 100 mV: $R_{shunt} \le 0.1\ V/0.1\ A$ = **1 Ω**.

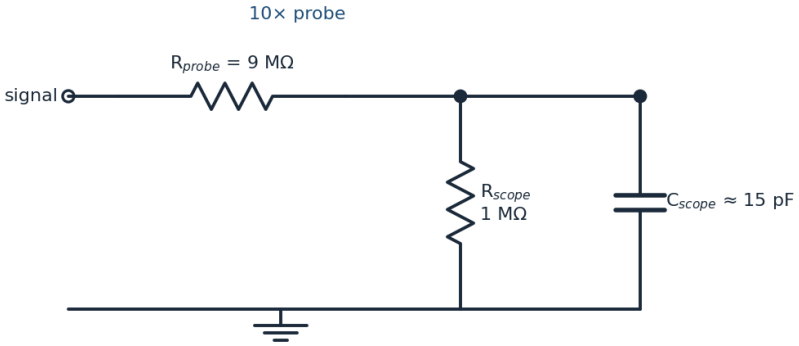
Step 3. With 1 Ω, 1.2 mA gives only 1.2 mV – hard to measure accurately with a normal DMM. Trade-off: small shunt = low burden but low signal. For µA–A ranges use an SMU or several switchable ranges.

Step 4. Power in the 1 Ω shunt at 100 mA = I^2R = 10 mW – negligible. Use a 4-wire (Kelvin) connection for low-ohm shunts.

Answer: 1.2 mA; use ≤ 1 Ω for a 100 mA range (burden ≤ 100 mV).

Problem 12.2 – Probe loading – 1× versus 10×

Question: A source with output resistance 100 kΩ is measured (a) by connecting it directly to a 1 MΩ scope input and (b) through a 10× probe (10 MΩ). Find the measurement error in each case.



Solution

Step 1. The source resistance and the scope input form a voltage divider.

Step 2. (a) Direct: V_{meas}/V_{true} = 1 MΩ/(1 MΩ + 0.1 MΩ) = 0.909 → error **−9.1 %**.

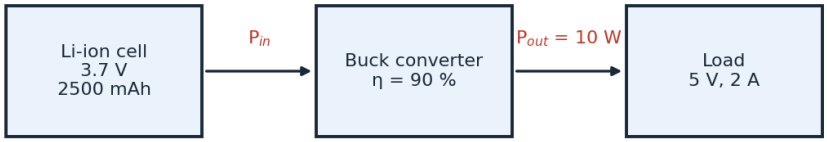
Step 3. (b) 10× probe: input resistance 10 MΩ → 10/(10 + 0.1) = 0.990 → error **−1 %**; the scope displays 1/10 of the value so set the channel to 10× (otherwise the reading is 10× too small).

Step 4. The probe also lowers the input capacitance from ≈ 100 pF (cable) to ≈ 10–15 pF, so fast signals are disturbed less. Always adjust the probe's compensation until a square wave is flat-topped.

Answer: (a) −9.1 %; (b) −1 %.

Problem 12.3 – Battery runtime with a DC-DC converter

Question: A 3.7 V, 2500 mAh Li-ion cell powers a 5 V, 2 A load through a buck-boost converter that is 90 % efficient. Find the input power, the battery current and the runtime.



Solution

Step 1. Output power = 5 V × 2 A = **10 W**.

Step 2. Input power = 10/0.9 = **11.1 W**.

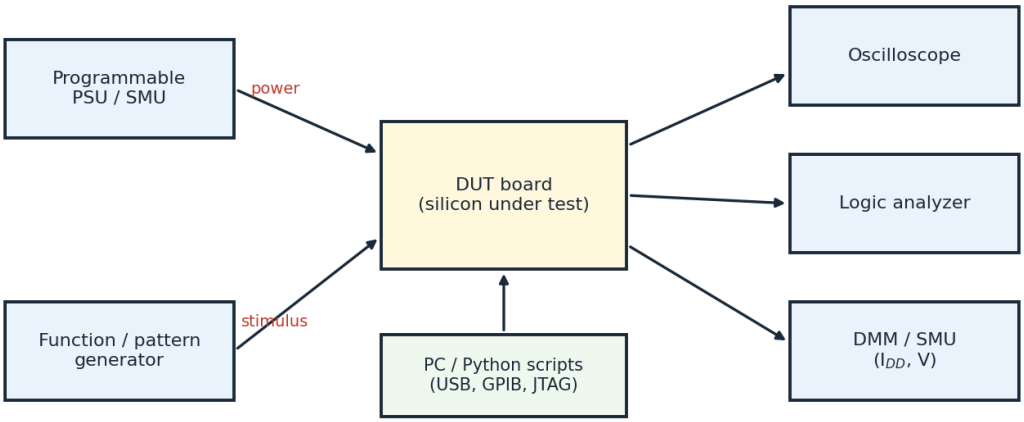
Step 3. Battery current = 11.1 W/3.7 V = **3.0 A** (1.2 C – acceptable but warm).

Step 4. Battery energy = 3.7 V × 2.5 Ah = 9.25 Wh → runtime = 9.25/11.1 = **0.83 h ≈ 50 minutes** (less in practice: aging, cold, cut-off voltage).

Answer: 11.1 W in, 3.0 A, about 50 minutes.

Problem 12.4 – Choosing the right instrument (bench set-up)

Question: For the bench set-up shown, choose the instrument for each task: (a) check supply ripple on the 1.2 V rail; (b) decode an SPI read; (c) measure sleep-mode current of 5 µA; (d) find the 3rd harmonic of a 100 MHz clock output.



Solution

Step 1. (a) **Oscilloscope** (AC-coupled, 20 MHz bandwidth limit off for fast spikes, short ground spring, 1× or low-capacitance probe).

Step 2. (b) **Logic analyzer** (or scope with serial decode) – shows SCLK, MOSI, MISO, CS together with decoded bytes.

Step 3. (c) **SMU / precision DMM in µA range** – a normal shunt would give an unreadable voltage; also disconnect other loads.

Step 4. (d) **Spectrum analyzer** – the harmonic is at 300 MHz, displayed in dBm/dBc.

Step 5. A PC with Python scripts (USB/GPIB/JTAG) automates all of it so the same test can be repeated over voltage and temperature.

Answer: (a) Scope, (b) Logic analyzer, (c) SMU, (d) Spectrum analyzer.

Quick concept questions

Q. Why are power-supply decoupling capacitors placed close to the IC?

A. To supply fast current spikes locally; trace inductance would otherwise create supply droop (V = L·di/dt).

Q. A 10× probe reads 200 mV and the scope channel is set to 1×. Actual signal?

A. The scope must be set to 10× – then the reading becomes **2 V**.

Q. A function generator set to 1 V (50 Ω source) drives a high-impedance scope input. What is displayed?

A. **2 V** – the generator assumes a 50 Ω load and shows half the open-circuit voltage.

13. Silicon Validation – Putting It All Together

In one line: silicon (post-silicon) validation proves that the real chip meets its specification across **PVT – Process, Voltage, Temperature**, then debugs every failure to its root cause.

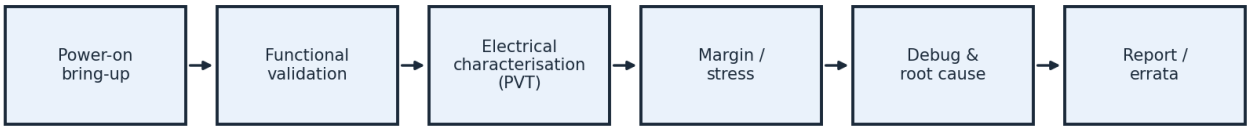


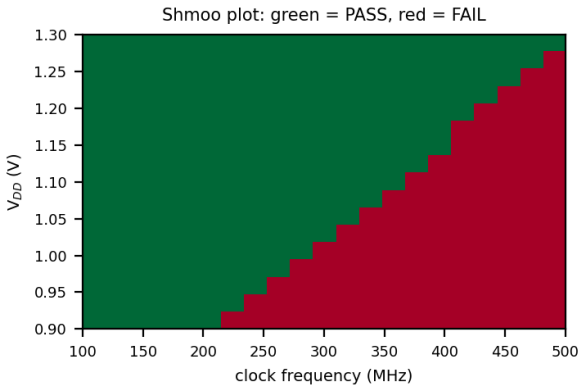
Fig 13.1 – Typical post-silicon validation flow.

Term	Meaning
Verification vs validation	**Verification**: pre-silicon simulation/emulation of the design against the spec. **Validation**: post-silicon testing of the real chip in the lab
Bring-up	First power-on: check rails, clocks/PLL lock, reset, JTAG access, then basic register read/write and boot
Characterisation	Measure parameters (V_{OH}/V_{OL}, I_{DD}, F_{max}, accuracy, INL/DNL, noise) across PVT and compare with datasheet min/typ/max
PVT corners	Process: slow/typical/fast silicon; Voltage: V_{nom} ± 10 %; Temperature: −40, 25, 85, 125 °C
Shmoo plot	Pass/fail map versus two parameters (e.g. frequency vs voltage)
Margining	Deliberately stress V, f, T or timing to measure how much margin exists
DFT / scan / JTAG / BIST	Design-for-test structures that give access to internal logic for debug and production test
ATE vs bench	ATE: production tester (fast, many devices). Bench: flexible lab set-up for debug
Errata	Documented confirmed silicon bugs and workarounds

13.1 Solved problems

Problem 13.1 – Reading a shmoo plot

Question: The shmoo plot shows where a chip passes (green) when the supply voltage and clock frequency are varied. (a) Roughly what is the maximum frequency at 1.0 V and at 1.2 V? (b) The product must run 300 MHz at 1.1 V ± 10 %. Is there margin? (c) What does the shape tell you?



Solution

Step 1. (a) At 1.0 V the green region ends near **≈ 290 MHz**; at 1.2 V it extends to **≈ 435 MHz** (read where the colour changes from green to red).

Step 2. (b) The lowest operating voltage is 1.1 × 0.9 = 0.99 V (≈ 1.0 V), where F_{max} ≈ 290 MHz < 300 MHz → the part **fails at the low-voltage corner**; it passes at nominal 1.1 V (≈ 360 MHz) and above.

Step 3. (c) F_{max} rises with voltage because transistors switch faster. The boundary is a timing-limited (setup) failure; a boundary that is vertical or irregular suggests a different cause (e.g. a hold or analog limit). Next steps: repeat at hot/cold and on several units, then locate the failing path using scan or ATPG data.

Answer: Fails at the 0.99 V corner (≈ 290 MHz < 300 MHz) – needs a design/voltage fix or a lower spec.

Problem 13.2 – Debug: works at 25 °C, fails at 125 °C

Question: A chip passes all tests at 25 °C but fails at 125 °C. Describe a systematic debug approach.

Solution

Step 1. Reproduce: confirm on several units and boards; rule out the test set-up (power, probe, socket, thermal soak time).

Step 2. Isolate the dependency: sweep voltage and frequency at 125 °C. If the failure goes away at a lower frequency or higher voltage → **timing** (hot silicon is slower; setup paths fail). If it depends on current → leakage/IR-drop.

Step 3. Check the physics: hot → lower mobility (slower), higher leakage (I_{DD} rises), analog references drift, oscillators/PLLs change range.

Step 4. Localise: use scan/BIST, on-chip monitors and a scope/logic analyzer to find the first failing block or path.

Step 5. Report: document the conditions, add the case to the test plan and, if confirmed as a design issue, record it as errata with a workaround.

Answer: Reproduce → vary V/f/T → identify the mechanism (timing, leakage, analog drift) → localise → document.

Quick concept questions

Q. How do you validate an ADC?

A. Static: slow ramp or histogram → offset, gain, INL, DNL. Dynamic: pure sine at several frequencies → FFT → SNR, THD, SFDR, ENOB. Repeat across PVT.

Q. Why terminate a fast signal with 50 Ω?

A. To match the transmission-line impedance and avoid reflections.

Q. What causes supply droop or ground bounce?

A. Simultaneous switching currents through package/board inductance ($V = L \cdot di/dt$); fix with decoupling capacitors close to the pins.

14. One-Page Formula Cheat Sheet

Topic	Essentials
R, L, C	$\tau_{RC} = RC$; $\tau_{RL} = L/R$; $f_c = 1/(2\pi RC)$; $X_C = 1/(2\pi fC)$; $X_L = 2\pi fL$; $f_0 = 1/(2\pi\sqrt{LC})$; $E_C = \frac{1}{2}CV^2$; $E_L = \frac{1}{2}LI^2$
Network	$V_{div} = V \cdot R_2/(R_1+R_2)$; $I_1 = I \cdot R_2/(R_1+R_2)$; $P_{max} = V_{th}^2/(4R_{th})$
Diode	$V_F \approx 0.7$ V; HW: $V_{dc} = V_m/\pi$; FW: $2V_m/\pi$; ripple $V_r = I/(fC)$; $R_s = (V_{in} - V_Z)/(I_Z + I_L)$
BJT	$I_C = \beta I_B$; $I_E = (\beta+1)I_B$; $\alpha = \beta/(\beta+1)$; $g_m = I_C/V_T$; $A_v(CE) = -g_m R_C$; CC gain ≈ 1
MOSFET	$I_D = \frac{1}{2}k(V_{GS} - V_{th})^2$ (sat, $V_{DS} \geq V_{ov}$); $g_m = 2I_D/V_{ov}$; $P_{dyn} = \alpha CV^2 f$
Op-amp	Inv: $-R_f/R_{in}$; Non-inv: $1 + R_f/R_g$; Integrator: $-(1/RC)\int$; GBW = const; $f_{max} = SR/(2\pi V_p)$; CMRR = $20 \log(A_d/A_{cm})$
Regulators	Linear: $P = (V_{in} - V_{out})I$; buck: DV_{in}; boost: $V_{in}/(1-D)$; LM317: $1.25(1 + R_2/R_1)$
Filters	-3 dB at f_c; -20 dB/decade per pole; $t_r = 0.35/BW = 2.2RC$; Sallen-Key $f_c = 1/(2\pi R\sqrt{(C_1C_2)})$
Digital	$n = \lceil \log_2 N \rceil$ FFs; Johnson = $2N$ states; ring = N; D: $Q^+ = D$; T: $Q^+ = T \oplus Q$; JK: $Q^+ = JQ' + K'Q$
Timing	$T \geq t_{cq} + t_{pd} + t_{su} + \text{skew}$; hold: $t_{cq,min} + t_{pd,min} \geq t_h + \text{skew}$
ADC / DAC	LSB = $V_{ref}/2^N$; SNR = $6.02N + 1.76$; ENOB = $(\text{SINAD} - 1.76)/6.02$; $f_s \geq 2f_{max}$; DNL = step/LSB $- 1$
Scope / power	BW $\geq 5f$; $t_r = 0.35/BW$; $t_{meas} = \sqrt{(t_s^2 + t_{scope}^2)}$; runtime = Ah/I; $\eta = P_{out}/P_{in}$
Interfaces	UART bytes/s = baud/10 (8N1); I²C $R_{p,max} = t_r/(0.8473C_b)$; SPI time = bits/$f_{clk}$
dB	20 log (voltage ratio) = 10 log (power ratio); -3 dB = 0.707; -6 dB = 0.5; $+20$ dB = ×10
C bit tricks	set: x |= 1u<<n; clear: x &= ~(1u<<n); toggle: x ^= 1u<<n; test: (x>>n)&1; clear lowest 1: x & (x−1)